

Duration and Street-Based Pricing for Incentive Compatible Paid On-Street Parking

(Authors' names are not included for peer review)

Abstract. *Problem definition:* Finding and paying for on-street parking remains a daily source of frustration for urban drivers, particularly in cities where advance payment is required and parking durations are uncertain. However, traditional pricing approaches often overlook behavioral factors that drive parking decisions. This study proposes an implementable pricing strategy that accounts for behavioral preferences to improve urban parking outcomes without relying on costly infrastructure. *Methodology/results:* Drawing on consumer behavior theory, we first treat parking spaces as substitutable products and use a choice-based conjoint survey to quantify how drivers respond to key parking attributes: price, walking distance to destination, and likelihood of extending parking time. Our findings from the survey emphasize that drivers' utility decreases as the levels of the parking attributes worsen. We then incorporate these behavioral estimates into a strategic pricing model that dynamically adjusts for demand shifts based on spatial and pricing factors. The optimization results show that (1) linear pricing leads to significant revenue loss, (2) both progressive and street-based pricing lead to higher revenue and occupancy, and their effectiveness depends on street characteristics and user behavior, and (3) the relative advantage of having a differentiated pricing policy compared to linear pricing peaks in medium duration and medium capacity environments where behavioral responsiveness and supply conditions are optimally aligned. *Managerial implications:* With this study, we showed that behaviorally grounded pricing strategies outperform traditional models assuming uniform rationality. For decision makers, this means that urban parking outcomes can be improved significantly without relying on new technologies by applying pricing strategies that reflect how drivers actually make trade-offs in practice.

Key words: pricing, optimization, consumer behavior, choice-based conjoint analysis

1. Introduction

Urban street parking management is a persistent and complex challenge for city planners. While alternative mobility options have expanded, many individuals remain heavily reliant on private vehicles, often due to perceived convenience and flexibility (Vega-Gonzalo et al. 2024). As cities continue to deal with the consequences of car dependency, the pressure on limited on-street parking spaces increases. This issue arises not only from lack of parking — drivers in NYC spend two hundred million hours a year circling the block searching for curbside parking, with typical searches lasting 20 minutes or more (Helfand 2025) — but also from inefficient pricing and management practices. For example, although demand for parking varies significantly throughout the week and across streets, one of the most commonly used pricing strategies, linear pricing, fails to account for these temporal and spatial dynamics (Onen Oz et al. 2025). As a result, cities struggle to balance

supply and demand effectively. Inspired by these observations and the gap in the literature, this study explores how implementing alternative pricing strategies can enable cities to better regulate parking demand across locations.

Dynamic pricing, the practice of adjusting parking fees in real time or at regular intervals based on demand conditions, is a promising strategy for parking management to balance supply and demand (Bayih and Tilahun 2024). Originally proposed by Vickrey (1954), dynamic pricing has only recently become feasible in practice thanks to advances in technology. A key technological development enabling real-time pricing adjustments is the use of ground sensor-based data, which allows municipalities to monitor occupancy levels and adapt prices accordingly. This initiative, nevertheless, has prohibitive drawbacks for municipalities, such as the high cost of installing and maintaining the systems (Zhao et al. 2021). As a cost-effective substitute, municipalities might utilize transaction data that tracks parking payments and provides valuable insights on usage patterns. However, transaction data reflects only realized demand -where drivers ultimately parked- rather than where they initially intended to park. Consequently, while ground sensors offer more accurate real-time data, neither method fully captures underlying parking preferences.

Relying solely on realized demand may lead to inefficient pricing by misinterpreting actual behavioral drivers. To address this, we model intended demand based on drivers' substitution behavior, recognizing that parking spaces function similarly to consumer products. As in many consumer product, since alternative parking options are not perfect substitutes (Kobus et al. 2013), drivers' choice may change based on price and convenience, which varies depending on both parking location and payment method. For instance, when on-street parking is unavailable or overly costly, some drivers might prefer to use off-street parking. Empirical research confirms this interdependence: as the price gap between off-street and on-street parking narrows, drivers engage in less cruising while searching for on-street spaces (Van Ommeren et al. 2012). Others, however, may continue circling the block for a cheaper spot or choose to park farther from their destination. This substitution behavior suggests that pricing changes at one location affect not only its own demand but also the demand for surrounding alternatives.

Given these dynamics, estimating intended demand requires accounting for both parking facility characteristics and drivers' substitution behavior. To accurately capture these, models must incorporate how factors such as price and convenience jointly influence choice. One key aspect of convenience is the likelihood that a driver will need to extend their parking duration. For cash and credit card users, the inability to extend remotely means they must either return to the parking

lot to extend their stay or risk receiving a fine if they overstay. This inconvenience is particularly problematic when the parking location is farther from their destination and when there is a high likelihood that additional parking time will be needed. In contrast, drivers using mobile payments can extend their parking remotely without interrupting their activities, eliminating the need to cover the distance again. However, while mobile payments reduce this inconvenience, the risk of receiving a fine still exists if drivers forget or are unable to extend on time; thus, the likelihood of extension remains a relevant cost factor for all users. These considerations suggest that distance, price, and extension-related uncertainty jointly influence drivers' parking choices. This leads to our first research question: *RQ1. How do drivers trade off Parking Price, Distance to Destination, and Likelihood of Extension when choosing between on-street and off-street parking options?*

Even though cities have begun to revise their pricing policies, many continue to rely on a constant hourly rate structure. However, as shown by Onen Oz et al. (2025), drivers' parking behavior is influenced by both price and its interaction with payment methods. Specifically, they showed that not only do payment durations vary based on drivers' payment method preferences, but also that the effect of price changes depending on the chosen payment method. Since convenience is not solely a function of proximity but also of the likelihood of extending the parking duration and the flexibility of doing so, a linear pricing structure may overlook important behavioral differences across driver segments. To address this, we evaluate two alternative pricing schemes, namely, linear street-based and progressive street-based pricing, and compare them to the conventional linear pricing model. Linear street-based pricing allows price differentiation across streets based on demand. Progressive street-based pricing adds another layer by increasing hourly rates with parking duration, with the aim of exposing drivers to different marginal costs depending on the length of their stay. These approaches aim to better align pricing with behavioral patterns and spatial demand variability. This leads to our second research question: *RQ2. How do progressive and street-based pricing strategies, compared to linear pricing, affect revenue generation and parking occupancy?*

In this study, we aim to systematically integrate driver preferences into pricing decisions to ensure that pricing strategies are aligned with user behavior and demand patterns. To find an answer to our first research question, we conduct an online survey where we use a between-subjects manipulation where we assign participants to different payment methods. With this survey, we measure driver preferences for different parking attributes through a choice-based conjoint (CBC) analysis that focuses on three attributes stated in our first research question: *Parking Price, Distance*

to Destination, and Likelihood of Extension. We use Hierarchical Bayes Multinomial Logit (HB-MNL) model to infer driver preferences. Furthermore, to address our second research question, we embed the estimated preferences into an optimization framework where we use a real-life transaction dataset to obtain the problem parameters. Drawing on Vulcano et al. (2012), which models primary demand for substitutable products, we define the probability of a driver choosing a specific parking space as a function of its attributes within the MNL framework. This allows us to move beyond traditional pricing models that assume static or location-specific demand, and instead capture substitution behavior across competing parking options. By embedding these behavioral predictions into our optimization model, we dynamically adjust prices to reflect their influence on intended demand, and propose a more responsive and behaviorally informed pricing strategy.

This study has several research contributions. First, we mimic real-life choice sets of drivers' to observe their substitution behavior. While doing so, different than the current literature, we focus not only on price but also on the convenience of drivers which might change depending on their preferred payment method. Our analysis shows that the utility of drivers decline monotonically with the increase in *Parking Price*, and we observe a nonlinear utility decline with the increase in *Distance to Destination* and *Likelihood of Extension*. These observations point to the presence of a psychological threshold where drivers change their behavior drastically with the attributes affecting their convenience. Second, we integrate predictive preference estimates along with real-life transaction data into a pricing optimization framework and aligns pricing decisions with observed driver behavior. Rather than treating demand as fixed or exogenous, the model uses estimated preferences to reflect how pricing decisions influence where drivers choose to park. This approach enables a more realistic and behaviorally grounded method for setting prices in urban parking systems. Our findings from the optimization model are threefold. First, municipalities currently relying on linear pricing are likely leaving substantial revenue on the table. Second, implementing street-based or progressive pricing policies offers clear advantages over linear pricing. Street-based pricing alone significantly increases revenue and occupancy by aligning prices with localized demand conditions. Combining it with progressive pricing further enhances both occupancy and revenue outcomes. Third, the effectiveness of differentiated pricing is influenced by both street-level characteristics and the behavioral profiles of users. The relative advantage of having a differentiated pricing policy compared to linear pricing peaks in medium duration and medium capacity environments where behavioral responsiveness and supply conditions are optimally aligned.

The remainder of the paper is structured as follows. The next section reviews relevant literature. Section 3 outlines the online survey design and presents the analysis of the collected data. Section 4 introduces the mathematical model developed to address the second research question. Section 5 details the computational studies and presents key findings along with managerial insights. The final section offers concluding remarks, discusses managerial implications, and outlines directions for future research.

2. Literature Review

This section examines two key research areas that are relevant to our investigation of parking pricing. We review each of these areas by summarizing relevant research findings and highlighting aspects where our research aims to contribute new insights.

2.1. Parking Choice

Recent research on parking demand has increasingly focused on understanding the factors that influence parking behavior and the selection of parking types. These studies employ various methodologies and consider a range of attributes affecting parking choices, as shown in Table 1.

Table 1 Literature Review on the Factors Affecting Parking Demand

	Main Attributes										
	Data	Model	PP	PD	OCC	TT	ST	DtD	PR	DEM	LoE
Hess and Polak (2009)	SP	ML	x			x	x	x	x	x	
Habib et al. (2012))	RP	DC	x			x			x	x	
Hilvert et al. (2012)	S&RP	ML	x	x		x	x	x	x	x	
Ibeas et al. (2014)	SP	ML	x				x	x	x	x	
Chaniotakis and Pel (2015)	SP	ML	x		x	x	x	x		x	
Soto et al. (2018)	SP	HDC	x				x	x		x	
Khaliq et al. (2019)	SP	ML	x	x				x			
Qin et al. (2020)	SP	MP	x		x		x	x	x	x	
Hassine et al. (2022)	RP	MNL	x	x		x	x	x	x	x	
Simićević and Milosavljević (2023)	S&RP	BL	x	x		x	x	x	x	x	
Rodríguez et al. (2023)	SP	ML	x	x		x	x	x	x	x	
<i>Our study</i>	<i>SP</i>	<i>MNL</i>	<i>x</i>	<i>x</i>				<i>x</i>			<i>x</i>

Note. SP, stated preferences; RP, revealed preferences; S&RP, combined stated and revealed preferences; MNL, multinomial logit; ML, mixed logit; DC, discrete-continuous; MP, mixed probit; HDC, hybrid discrete choice; BL, binary logit; PP, parking price; PD, parking duration; OCC, occupancy; TT, trip time; ST, search time; DtD, distance to destination; PR, purpose of trip; DEM, demographics; LoE, Likelihood of Extension.

Several studies rely on stated preference (SP) data, where participants provide hypothetical choices in controlled scenarios (Hess and Polak 2009, Ibeas et al. 2014, Chaniotakis and Pel 2015, Soto et al. 2018, Khaliq et al. 2019, Qin et al. 2020, Rodríguez et al. 2023). SP data allows researchers to design experiments that test preferences for hypothetical scenarios, enabling them to evaluate factors that may not yet exist in the real world. However, these responses may be subject

to hypothetical bias as individuals do not always behave as they claim they would. In contrast, revealed preference (RP) studies analyze actual parking behaviors based on real-world observations (Habib et al. 2012, Hassine et al. 2022). RP data provides insights into real decisions made under actual constraints but are limited by the availability of observed variation in parking conditions. Some studies incorporate both SP and RP data to improve model robustness (Hilvert et al. 2012, Simićević and Milosavljević 2023), combining the benefits of controlled experimental design with real-world validation.

Key factors affecting parking demand include parking price (PP), parking duration (PD), occupancy (OCC), trip time (TT), search time (ST), distance to destination (DtD), purpose of trip (PR), and demographic characteristics (DEM). The majority of studies consider multiple attributes, with parking price (PP) being the most frequently examined factor. Other attributes, such as trip time (TT) and demographics (DEM), are also extensively analyzed, while occupancy (OCC) and search time (ST) receive relatively less attention.

As there are numerous contributing factors the importance of which is still being researched and the dependent variable (parking type choice) is categorical, authors typically apply discrete choice models. Mixed logit (ML) models are commonly used due to their flexibility in capturing unobserved heterogeneity (Hess and Polak 2009, Hilvert et al. 2012, Ibeas et al. 2014, Chaniotakis and Pel 2015, Khaliq et al. 2019, Rodríguez et al. 2023). Other models, such as discrete-continuous (DC) (Habib et al. 2012), hybrid discrete choice (HDC) (Soto et al. 2018), mixed probit (MP) (Qin et al. 2020), multinomial logit (MNL) (Hassine et al. 2022), and binary logit (BL) (Simićević and Milosavljević 2023), have been applied to account for different aspects of parking behavior.

While existing research identifies various factors that shape parking choices, such as search time, perceived safety, and environmental conditions, operationalizing these in optimization models often requires real-time data or strong behavioral assumptions. Moreover, findings across studies vary significantly based on local context and user habits. To address these challenges and ensure practical applicability, we focus on three core attributes that are central to driver decision-making and align with our transaction data: *Parking Price*, *Distance to Destination*, and *Likelihood of Extension*. Among these, the introduction of the *Likelihood of Extension* as a choice attribute represents a novel contribution. While prior studies have primarily focused on price and spatial proximity, they have largely overlooked the behavioral uncertainty drivers face regarding the ability to extend their parking session. This omission is critical, as the risk of enforcement penalties or the inconvenience of having to relocate can significantly change parking preferences. By explicitly incorporating

the *Likelihood of Extension*, our study captures an important dimension of user behavior that influences location choice. Additionally, by incorporating payment method and controlling for expected duration, we account for contextual variability in driver behavior.

2.2. Parking Pricing

As urban mobility patterns evolve, parking demand fluctuates based on various factors, such as time of day, location, and external events. To respond to these variations efficiently, it is believed that adapting dynamic pricing policies that adjust parking prices in real time might be beneficial to optimize occupancy and revenue. However, these pricing strategies are not implemented in isolation. Different stakeholders, such as parking garage operators and municipal or on-street parking authorities, compete for drivers by setting optimal prices, taking into account the responses of their customers. These drivers, in turn, make decisions based on multiple factors, including proximity to destinations, pricing, and congestion levels (Bayih and Tilahun 2024).

This interaction between pricing policies and user decisions highlights the need for models that capture the dynamic relationship between agencies and drivers. Bi-level and game theoretical optimization problems are among the methods used to include this decision-making interaction. For example, Mackowski et al. (2015) established a dynamic non-cooperative bi-level model to obtain real-time parking prices. Numerical examples showed that the pricing model can eliminate cruising for parking, balance parking space utilization, and reduce traffic congestion. Lei and Ouyang (2017) used a multi-period non-cooperative bi-level model to determine the location-dependent parking prices. The results indicated that a pricing policy combined with parking supply and demand information can improve the performance of the parking system. Building on the work by Mackowski et al. (2015), Mirheli and Hajibabai (2019) developed a dynamic parking pricing and parking utilization model in which uncertainty in future demand and occupancy at each facility is addressed. To solve their model, a Monte Carlo tree search heuristic is used. Fotouhi and Miller-Hooks (2021) extended Mackowski et al. (2015)'s single-owner problem to a multi-owner and competitive problem setting. Their results showed that increasing crowdsourced parking spaces and competitive pricing benefits the parking market's social welfare. Recently, Bayih and Tilahun (2025) developed a bi-level optimization model to capture the interaction between parking demand, pricing strategies, and capacities. Their findings demonstrated that dynamic pricing can effectively align parking supply with demand, which leads to improved revenue outcomes for parking operators.

Despite the appeal of bi-level and game-theoretic models in integrating hierarchical decision-making, their model complexity scale up quickly, which limits their applicability to large-scale urban

environments. Although linear pricing is easy to implement and still helps changing parking demand (Simićević et al. 2012, Nourinejad and Roorda 2017, Khordagui 2019), occupancy (Chatman and Manville 2014, Mo et al. 2021) and search time (Alemi et al. 2018, Najmi et al. 2021), they often overlook the role of payment methods in shaping pricing effectiveness. As parking duration of drivers preferring different payment methods is different, utilizing a progressive pricing structure will provide the additional benefits of market segmentation (Shoup 2021). Instead of imposing higher costs on drivers parking during peak demand periods, progressive pricing charges higher rates based on parking duration. This approach incentivizes shorter stays, improves parking turnover, and enhances availability. Despite its advantages, progressive pricing remains underexplored in academic literature. Although implemented in some urban areas, its broader adoption and systematic analysis remain limited. Marcucci et al. (2015) took a qualitative approach to argue that progressive pricing can increase parking turnover and improve availability, while generating higher revenues. Ornelas et al. (2023) contributed to the literature on progressive pricing by formulating driver reactions to this policy and modeling its optimal profile design. Their simulation results showed that progressive pricing outperforms linear pricing in terms of both revenue and occupancy outcomes.

Our study contributes to this literature by offering a systematic, data-driven evaluation of progressive pricing using a combination of survey-based preference estimation and optimization modeling. Moreover, existing studies often treat pricing structure independently of payment methods, despite evidence that drivers using different payment methods exhibit distinct duration patterns and price sensitivities. By explicitly incorporating payment method as a between-subjects factor in a choice-based conjoint experiment, and integrating the resulting behavioral insights into a parking price optimization model, our study highlights the operational benefits of aligning pricing strategies with user segmentation. This approach allows us to evaluate the revenue and occupancy effects of progressive pricing not in isolation, but in comparison to both linear and street-based pricing.

3. Survey Design and Results

To explore how various parking attributes influence drivers' choices regarding parking alternatives, we have created an online survey that examines participants' decision-making processes in different scenarios. This section will first introduce our online survey along with its methodology. Then, we will discuss the results that may assist municipalities in enhancing street parking.

3.1. Survey Design

The goal of our survey is to collect choice data to measure consumer preferences for parking attributes using choice-based conjoint (CBC) analysis to find answers to our first research question.

In CBC, respondents are repeatedly presented with varying subsets of possible parking scenarios and at each choice task are asked to select the most preferred option. In this context, we leverage a design with three choice alternatives plus a “no-choice” option. Each alternative is a combination of levels for a limited set of attributes. We describe each parking alternative using three attributes. In designing our survey for analyzing parking behavior, we carefully balance the number of tasks, the complexity of attributes, and the efficiency of the design.

Attributes and Levels. There are two key decisions when designing a survey. The first one is regarding to the attribute selection. Green and Srinivasan (1990) suggested using fewer than seven attributes to maintain simplicity and avoid overwhelming respondents. In accordance with the suggestion from the literature, for attribute selection in our survey, we focused on three attributes that are important in parking decision-making. To effectively model driver behavior, we included *Parking Price* and *Distance to Destination*, since these are known factors of parking decision-making. *Parking Price* directly affects demand elasticity, while *Distance to Destination* represents the trade-off between cost and convenience. In addition to these, we incorporated *Likelihood of Extension*, which captures the probability that a driver will need additional parking time. This attribute is crucial as drivers who expect a greater extension likelihood may be more cautious in their initial parking choices. The impact of this uncertainty is further shaped by the convenience of different payment methods, since drivers who pay with cash or credit cards must physically return to extend their parking, whereas those using mobile payments can do so remotely. Thus, drivers who are likely to require additional time may consider not only price and proximity but also the ease of extending their stay when making their initial decision.

The second decision is about determining how many levels to include for each attribute. Since the findings from this survey will be used as parameters in our optimization model, we prioritized aligning the survey design with the real-world transaction data collected by our partner organization in one of North America’s most densely populated urban centers. To do so, we determined the appropriate number of levels for *Likelihood of Extension* by deriving an approximation from the transaction data. As transaction data does not directly record whether a driver extended their parking duration, we inferred extensions using two distinct approximation methods. In the first one, we assumed that streets where occupancy fluctuated more suggested that drivers were less predictable and more likely to extend their parking durations. To capture this, we calculated the coefficient of variation (CV) for each street’s occupancy, which captures the extent of variability relative to the mean occupancy. In our second approximation, we identified likely extensions based

on short gaps between consecutive transactions at the same parking spot, and we calculated the probability of extension for each street. We then clustered streets using k-means clustering. For both approximations, the elbow method suggested using five clusters for best representation of variability patterns. Based on this analysis, we defined five levels for the *Likelihood of Extension* attribute. To avoid the “number-of-levels” effect, where attributes with more levels tend to appear more important (Currim et al. 1981, Verlegh et al. 2002), we used five levels for all attributes. This choice also follows the guidance from Eggers et al. (2021), who recommended keeping the number of levels for each attribute limited to avoid exponential increases in experimental design complexity. Table 2 shows the attributes used in our study with their corresponding levels.

Table 2 Attributes and Their Corresponding Levels

Attributes	Levels	Values
Parking Price	5	3 \$/hour, 4 \$/hour, 5 \$/hour, 6 \$/hour, 7 \$/hour
Distance to Destination	5	50m, 100m, 200m, 400m, 800m
Likelihood of Extension	5	<20%, 20%-40%, 41%-60%, 61%-80%, >80%

We also considered two other attributes that might affect the parking choice of drivers. We assigned participants to one of three experimental conditions, each corresponding to a different payment method: cash, credit card, or mobile application. In our survey design, we deliberately chose not to include payment method as one of the attributes varied within the choice tasks. We discussed the reasons behind this design choice in Section EC.1.1 in detail. Furthermore, the survey incorporated five expected parking durations, which were displayed to participants at the top of the screen during the CBC tasks. Each participant saw all five duration ranges exactly four times, with the order of presentation balanced across participants using Latin square design. For further information, please refer to Section EC.1.3.

Choice Set Design. In our survey, we employed three choices per task and a “no-choice” to overcome the limitations associated with using pairs, as highlighted by Meissner et al. (2016). Each participant responded to 20 choice tasks in accordance with the recommendation of Johnson and Orme (1996). We generated our choice tasks using the *%MktEx* and *%Choiceff* macros in SAS. We provide further details about the survey in Section EC.1.

3.2. Survey Results

We estimate individual-level part-worth utilities using a Hierarchical Bayesian Multinomial Logit (HB-MNL) model. Each respondent $i \in \{1, \dots, I\}$ faces a series of K choice tasks, in which they

are asked to select one of N available alternatives, including a “no-choice” option. Each alternative is defined by a combination of attribute levels, and we assume that choices are modeled as the outcome of a utility-maximizing process subject to random shocks. The utility that respondent i derives from alternative $j \in \{1, \dots, N\}$ in task k is given by:

$$U_{ijk} = \sum_{a=1}^A \sum_{l=1}^{L_a-1} \beta_{ial} x_{ijkal} + \varepsilon_{ijk}.$$

Here, x_{ijkal} denotes the effects-coded design matrix corresponding to attribute a and level l for participant i , task k , and alternative j , and β_{ial} corresponds to the part-worth coefficient of participant i for level l of attribute a . We used effects coding for all categorical attributes to avoid arbitrary baseline choices and to facilitate interpretation of each level’s utility relative to the attribute’s grand mean. We represented each categorical attribute using $L_a - 1$ binary variables. For each attribute a , the part-worth utility of the last level is calculated as $\beta_{iaL_a} = -\sum_{l=1}^{L_a-1} \beta_{ial}$. Finally, the random error term ε_{ijk} is assumed to follow a Gumbel(0,1) distribution.

We assume that the individual-level part-worth utilities, β_i , to follow a non-centered hierarchical structure:

$$\beta_i = \left[\beta_{i0}, \beta_{i11}, \dots, \beta_{iA(L_a-1)} \right]^T \sim \text{MVN}(\Theta \mathbf{z}_i, \Sigma),$$

where Σ represents a diagonal covariance matrix. The vector $\mathbf{z}_i$ captures participant-level covariates that may influence preferences. Specifically, we included variables such as age, income, sex, and living area. Additionally, we incorporated responses to recency, frequency, monetary value questions to account for differences in parking behavior and purchasing tendencies of participants. These covariates helped us explain between-participant variation in part-worth utilities and allowed the model to generalize more effectively. However, we did not include expected parking duration and payment method as attributes in the estimation model, as participants did not choose them within tasks. As previously explained, expected parking duration was experimentally varied across blocks of tasks using a Latin square design, while payment method was designed as a between-subject condition. Since both factors remained constant within tasks, we treated them as contextual variables and examined their influence on choice behavior through post hoc analyses.

Estimation Approach. We estimate the parameters for the above HB-MNL model with Hamiltonian Monte Carlo in Python STAN. Details about the model is given in Section EC.1.5. We run the model with 2,000 warm-up iterations and 4,000 sampling iterations per chain across four chains, yielding a total of 16,000 post-warm-up draws for posterior inference. To ensure robust

model validation, we implement 5-fold validation using five versions of the design to assess model performance across multiple respondent subsets. Unlike traditional holdout methods, which reuse participants for both estimation and validation and test only at the question level, k-fold validation rotates holdout samples. This reduces performance variance and avoids overfitting to specific respondents, and lead to more reliable out-of-sample predictions Elrod (2001). Table EC.3 shows that our model has both high in-sample and out-of sample fit, and conclude the model fit is adequate. Furthermore, all standard diagnostic checks, namely, divergent transitions, tree depth, effective sample size, and $\hat{R}$ values, as discussed in Section EC.1.6, indicate that the model has converged appropriately and has yielded reliable estimates.

Part-worth Utilities. To achieve reasonable statistical power, we recruited 450 participants from Prolific, paying each £2.7 as compensation. We removed three participants from the analysis as they selected “no-choice” options across all 20 choice tasks. We also repeated all analyses using only participants who answered both attention check questions correctly, and additionally excluded those who had never used paid on-street parking; our results remained robust (Table EC.4).

Figure 1 visualizes the estimated part-worth utilities and their associated 95% confidence intervals for each level of the three attributes, while Table 3 presents the corresponding numerical estimates. Since effects coding centers each attribute around its mean, the utility estimates represent the deviation of each level from the average desirability for that attribute. It is also worth noting that since the fifth level of each attribute (\$7 for *Price*, 800 meters for *Distance to Destination*, and a *Likelihood of Extension* above 80%) serves as the baseline in the effects-coded design, it is not shown explicitly in Figure 1 and Table 3. However, its utility can be inferred as the negative sum of the estimated utilities for the other levels within that attribute.

Figure 1 Posterior Part-worth Utility Distributions

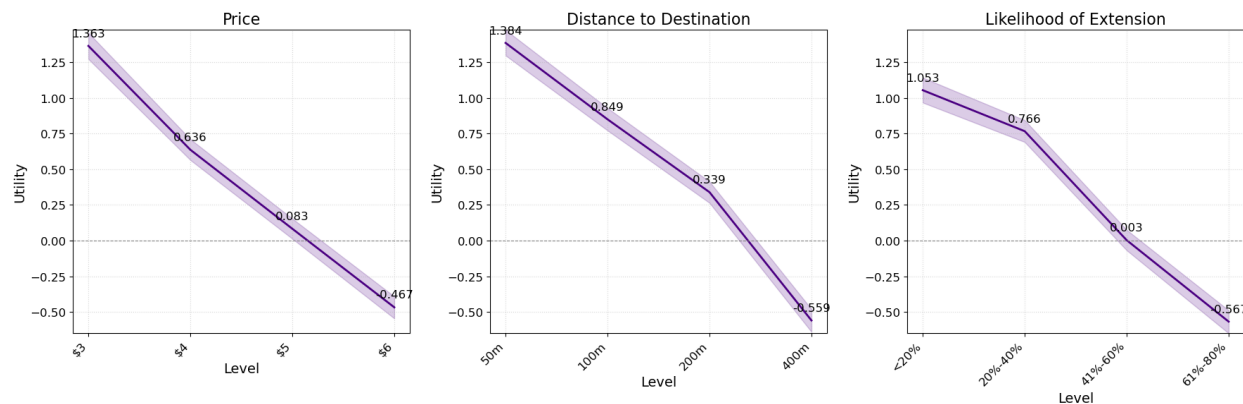


Table 3 Posterior Part-worth Utility Estimates

Attribute Label	Mean	Std. Dev.	Lower 95% CI	Upper 95% CI
$\beta_{NoChoice}$	-3.648	0.232	-4.132	-3.226
$\beta_{Price:3\$/hour}$	1.363	0.047	1.271	1.455
$\beta_{Price:4\$/hour}$	0.636	0.037	0.564	0.710
$\beta_{Price:5\$/hour}$	0.083	0.036	0.013	0.154
$\beta_{Price:6\$/hour}$	-0.467	0.040	-0.545	-0.391
$\beta_{Distance:50m}$	1.384	0.045	1.296	1.475
$\beta_{Distance:100m}$	0.849	0.040	0.771	0.928
$\beta_{Distance:200m}$	0.339	0.038	0.264	0.413
$\beta_{Distance:400m}$	-0.559	0.041	-0.639	-0.480
$\beta_{Extension:<20\%}$	1.053	0.045	0.966	1.141
$\beta_{Extension:20\%-40\%}$	0.766	0.039	0.690	0.843
$\beta_{Extension:41\%-60\%}$	0.003	0.036	-0.068	0.075
$\beta_{Extension:61\%-80\%}$	-0.567	0.041	-0.649	-0.488

For the *Parking Price* attribute, utility estimates decline monotonically, ranging from 1.363 (95% CI: [1.271, 1.455]) for \$3 to -0.467 (95% CI: [-0.545 , -0.391]) for \$6. This nearly linear pattern suggests that participants evaluated price in a proportionate manner, with each dollar increase generating a relatively consistent drop in utility. The slope is slightly steeper at lower price levels, which implies that participants are particularly sensitive to changes in lower price ranges.

In contrast, the *Distance to Destination* attribute shows a nonlinear utility decline as walking distance increases. The utility for very close parking (50 meters) is 1.384 (95% CI: [1.296, 1.475]), which drops to 0.849 for close distance (100 meters), 0.339 for moderate distance (200 meters), and -0.559 (95% CI: [-0.639 , -0.480]) for far distance (400 meters). These estimates imply that while participants are tolerant of short and moderate walking distances, their utility declines more sharply beyond 200 meters.

The rightmost panel displays the utilities for different levels of *Likelihood of Extension*. This attribute follows a similarly decreasing pattern, with lower extension likelihoods being more desirable. A likelihood of less than 20% yields a utility of 1.053 (95% CI: [0.966, 1.141]), decreasing to 0.766 for 20–40%, then sharply dropping to nearly zero for 41–60% (mean: 0.003, 95% CI: [-0.068 , 0.075]), and further to -0.567 (95% CI: [-0.649 , -0.488]) for 61–80%. Unlike the relatively linear pattern for *Parking Price*, the shape of the curve suggests that participants are relatively tolerant of lower extension risks but exhibit a sharper decline in preference once uncertainty about parking duration increases beyond 40–60%. This again points to the presence of a psychological threshold as in the case of *Distance to Destination*, where perceived uncertainty becomes a significant deterrent.

As a post hoc analysis, we looked into the differences between utilities across different payment methods. While the overall preference patterns were similar, we observed meaningful differences

in how participants evaluated *Distance to Destination* and the *Likelihood of Extension*. Participants assigned to cash or credit card exhibited greater sensitivity to unfavorable levels of both attributes, as reflected in steeper declines in utility. In contrast, participants assigned to mobile application were less negatively affected by increased distance or a higher extension likelihood. Our explanation is that mobile applications make it more convenient to manage parking remotely; therefore, it reduces the impact of uncertainty related to duration or inconvenience of walking longer distances. These differences in utility slopes hint at varying sensitivities across groups. We further quantified these differences using normalized attribute importance scores in Section EC.1.8.

Furthermore, when we examined utility estimates across expected parking durations, we found that *Parking Price* sensitivity increased with duration. That means, participants placed more weight on price differences when they expected to park longer. Additionally, sensitivity to *Distance to Destination* is fairly consistent but peaks slightly at mid-to-long durations. Specifically, when drivers anticipate longer stays, they become more price-sensitive and more concerned about distance to their destination, likely because the cumulative cost and inconvenience grow with time. In contrast, sensitivity to the *Likelihood of Extension* remained relatively stable, which indicates that time uncertainty was generally seen as undesirable, regardless of initially planned to parking duration.

4. Model Framework

Our survey data provided insights into the factors affecting drivers' preferences on their parking choice. In this section, we build on these findings and introduce our mathematical model to be used in finding optimal parking prices.

We describe the transportation network as a grid $\Gamma(N, A)$ where N shows the set of streets and A corresponds to a directed link between two streets with no intersections. We consider Manhattan distance between streets. Specifically, let d_{ij} show the distance between streets i at (x_i, y_i) and j at (x_j, y_j) , where $d_{ij} = |x_i - x_j| + |y_i - y_j|$.

Driver Characteristics and Behavior. A driver arrives on the grid with a most preferred street, which typically the one closest to their destination, where they would ideally like to park. We define a driver's type based on three characteristics: (i) their most preferred street i , (ii) their required parking duration, denoted by v , which can be either *short* (S) or *long* (L), where the mean duration in terms of periods is μ_{iS} and μ_{iL} , respectively, and (iii) their preferred payment method m . We denote $\kappa_i \in [0, 1]$ as the probability that a driver from street i is a short-term (type S) driver, and $1 - \kappa_i$ as the probability that the driver is a long-term (type L) driver. Furthermore, probability of selecting payment method m on street i will be denoted by $\delta_{mi} \in [0, 1]$.

Payment Duration Selection. Drivers generally do not know in advance whether they will end up staying for a short or long term. Therefore, they must choose a payment option under uncertainty. A driver compares the expected regret from paying for a short duration versus a long duration and chooses between the two probabilistically, assigning higher likelihood to the option that yields lower regret.

If a driver pays for a short duration (S), and their actual stay is indeed short (S), no regret is incurred. However, if their actual stay turns out to be long (L), they may need to extend the parking session. The inconvenience of this extension is captured by a hassle cost h_m , which depends on the payment method m : mobile users can increase their payment remotely, while cash or card users must return to the vehicle. Hence, the expected regret associated with initially paying for a short duration is:

$$\text{Regret}_S = \kappa_i 0 + (1 - \kappa_i) h_m = (1 - \kappa_i) h_m.$$

On the other hand, if a driver pays for a long duration (L), a long-type driver incurs no regret. However, if the driver is actually a short-type driver, they overpay for unused parking time. The regret in this case is quantified as the monetary cost of the unused portion of time, calculated as $(\mu_{iL} - \mu_{iS}) p_j^L$, where p_j^L is the per-unit parking price on street j for long-term parking. The expected regret associated with initially paying for a long duration is then:

$$\text{Regret}_L = \kappa_i (\mu_{iL} - \mu_{iS}) p_j^L.$$

To translate these regret values into choice probabilities, we adopt a regret-based probabilistic decision rule. Rather than assuming that all drivers deterministically select the lowest-regret option, we allow for bounded rationality by modeling choice behavior as probabilistic, where options with lower expected regret are more likely to be selected, but not guaranteed. Specifically, we define the probability of paying for a short duration as the normalized inverse of its associated regret:

$$\rho_{vS}^{ijm} = \frac{\text{Regret}_L}{\text{Regret}_S + \text{Regret}_L} = \frac{\kappa_i (\mu_{iL} - \mu_{iS}) p_j^L}{(1 - \kappa_i) h_m + \kappa_i (\mu_{iL} - \mu_{iS}) p_j^L}. \quad (1)$$

The probability of paying for a long duration is simply the complement:

$$\rho_{vL}^{ijm} = 1 - \rho_{vS}^{ijm}. \quad (2)$$

A distinguishing feature of our model is that these $\rho_{v\phi}^{ijm}$ values are not assumed a priori; rather, they are endogenously determined based on the expected regret associated with each payment

choice and parking price. These probabilities reflect drivers' decision-making under uncertainty about their actual parking duration.

Parking Choice Behavior. We denote the baseline demand by N_{it} , which represents the number of drivers who prefer street i at time t . However, instead of assuming that all of these drivers will attempt to park on their preferred street, we model their destination choices probabilistically using the MNL framework. The MNL structure captures drivers' substitution behavior across all available streets based on three key attributes: the parking price on street j for τ duration (p_j^τ), the distance from the preferred street to the street j (d_{ij}), and the estimated likelihood of needing to extend the parking session on street j (le_{ij}). Each of these attributes contributes to the deterministic part of the utility for parking on street j . To reflect preference heterogeneity and bounded rationality in real-world behavior, we simulate the full utility by adding a Gumbel-distributed random noise term ($\varepsilon_{ij} \sim \text{Gumbel}(0, 1)$) to each street's utility consistent with the HB-MNL model we used to estimate part-worth utilities in our survey analysis. We also introduce an outside option to represent the case where a driver decides not to park on any of the available streets. Its utility is similarly modeled as the sum of a baseline utility corresponding to the "no-choice" option and a Gumbel-distributed error term. Let ω_{ij} denote the probability that a driver who prefers street i chooses to attempt parking on street j , and let ω_{iJ} denote the probability of opting out. Then:

$$\omega_{ij} = \frac{e^{\eta_{ij}}}{e^{\eta_{iJ}} + \sum_{k \in N} e^{\eta_{ik}}}, \quad \forall i, j \in N \cup J, \quad (3)$$

where

$$\eta_{ij} = \beta_0 + \beta_1 \frac{p_j^S + p_j^L}{2} + \beta_2 d_{ij} + \beta_3 le_{ij} + \varepsilon_{ij}. \quad (4)$$

In defining the utility of drivers, we use both short- and long-term parking prices. This specification reflects the driver's exposure to both pricing rates in forming price expectations, especially when the duration choice is not yet finalized.

We model destination choice using a classical MNL structure, where the probability of a driver attempting to park on a particular street depends on observed and anticipated street-level attributes. Although real-time availability may play a role in parking decisions, we do not incorporate it directly into the utility function. This modeling choice is supported by two factors. First, urban drivers consider only a limited number of visibly distinct alternatives, which makes the main assumption of the MNL model less problematic and allows the MNL model to be a reasonable representation of substitution behavior. Second, in practice, most drivers do not access real-time occupancy data

and instead rely on prior experience or heuristics to anticipate availability. To reflect this, we treat availability as an external constraint enforced during the assignment phase, rather than a dynamic factor influencing utility directly.

The likelihood that a driver will need to extend their parking session used in the utility expression for ω_{ij} depends on the probability that a long-term driver (L) initially pays for a short duration (S). This is the only case in which a parking extension becomes necessary. Therefore, likelihood of extension is captured by:

$$le_{ij} = \sum_{m \in M} \rho_{LS}^{ijm},$$

where ρ_{LS}^{ijm} is defined as the probability that a driver who turns out to be of long type selects to pay for a short duration, given method m .

Effective Demand. We compute the *effective demand*, defined as the portion of baseline demand N_{it} that proceeds to attempt parking, by scaling N_{it} with the probability of not choosing the outside option:

$$\tilde{N}_{it} = N_{it}(1 - \omega_{iJ}). \quad (5)$$

This adjustment introduces price sensitivity into the model. When parking prices are high or attributes are unfavorable, more drivers choose to opt out, and, consequently, the effective demand decreases. However, we do not interpret $\tilde{N}_{it}$ as a fixed quantity to be partitioned across disjoint actions, but rather as the baseline mass of arriving drivers from which multiple probabilistic behavioral outcomes are generated.

Let $y_{ijtm}^{v\phi}$ denote the expected number of type- ivm drivers who attempt to park on street j in period t , where $\phi \in \{S, L\}$ indicates the paid duration. This value depends on the total number of drivers of type i at time t , $\tilde{N}_{it}$, and we assume that drivers select a particular combination of street, payment method, and paid duration, conditional on their type and the regret-minimizing structure of the model. Specifically, we express $y_{ijtm}^{v\phi}$ as:

$$y_{ijtm}^{v\phi} = \begin{cases} \delta_{mi} \tilde{N}_{it} \omega_{ij} \rho_{v\phi}^{ijm} \kappa_i, & \text{if } v = S, \phi = S, L, \\ \delta_{mi} \tilde{N}_{it} \omega_{ij} \rho_{v\phi}^{ijm} (1 - \kappa_i), & \text{if } v = L, \phi = S, L. \end{cases} \quad (6)$$

This expression considers drivers' destination preferences, captured by the choice probability ω_{ij} , probability of selecting payment method m on street i , $\delta_{mi} \in [0, 1]$, and the likelihood that a driver of true duration type $v \in \{S, L\}$ chooses to pay for a duration of type $\phi \in \{S, L\}$ on street j , given payment method m , $\rho_{v\phi}^{ijm}$. **[This model assumes that location and payment method choices**

as independent decisions, since parking location is typically constrained by spatial and temporal factors, while payment method depends on infrastructure and user habits. Moreover, we account for the differences between payment methods indirectly by modeling drivers' expectations about extension flexibility through le_{ij} .]

Capacity Usage. We can define the post-assignment usage of parking space in each street. Due to limited parking capacity on each street, not all drivers who wish to park can be accommodated. Therefore, we introduce a decision variable $a_{ijtm}^{v\phi}$ to represent the number of type- ivm drivers who are actually assigned a parking space on street j in period t . The variable $a_{ijtm}^{v\phi}$ is bounded above by $y_{ijtm}^{v\phi}$, and allocation is subject to parking capacity constraints over time, where we divide the day into 15-minutes time intervals $t \in \{1, \dots, T\}$ for tractability. Specifically, the number of spots that are in use on street j at time t can be defined as;

$$\mathbb{L}_{jt} = \mathbb{L}_{jt-1} + \sum_{m \in M} \sum_{i=1}^N \sum_{v, \phi \in \{S, L\}} a_{ijtm}^{v\phi} - \sum_{m \in M} \sum_{i=1}^N \sum_{\phi \in \{S, L\}} a_{ij(t-\mu_{iL})m}^{L\phi} + a_{ij(t-\mu_{i\phi})m}^{S\phi}$$

The binary variables s_{jt} are defined to denote if there is available capacity on street j at time t :

$$s_{jt} \leq Cap_j - \mathbb{L}_{jt}, \quad \forall j \in N, \forall t \in T, \quad (7)$$

$$Cap_j - \mathbb{L}_{jt} \leq s_{jt} Cap_j, \quad \forall j \in N, \forall t \in T, \quad (8)$$

$$\mathbb{L}_{jt} \leq Cap_j, \quad \forall j \in N, \forall t \in T,$$

$$s_{jt} \in \{0, 1\}, \quad \forall j \in N, \forall t \in T.$$

Pricing Decision and the Objective Function. Given the capacity Cap_j of street $j \in \{1, \dots, N\}$, the municipality must determine the parking price. In most cities, municipalities apply a constant hourly parking rate per street independent of the duration selected by drivers. As an alternative to this pricing model, progressive pricing policies can be implemented, where the hourly rate changes after a certain duration threshold to discourage long stays and promote turnover. To reflect this pricing structure, our model introduces two separate pricing levels: one for short durations and another for long durations, p_j^τ for each street j for duration $\tau \in \{S, L\}$. This enables a more accurate representation of how price sensitivity affects duration decisions. In doing so, the model provides a richer foundation for exploring pricing interventions and their impact on both user behavior and municipal revenue. Each price is bounded by a lower and upper limit through the constraint:

$$lb \leq p_j^\tau \leq ub, \quad \forall j \in N, \forall \tau \in \{S, L\}. \quad (9)$$

To compare this progressive street-based pricing approach with traditional linear street-based pricing, we include a constraint enforcing $p_j^S = p_j^L$ to effectively reduce the model to the linear street-based pricing case. Going further, we can eliminate both duration- and street-level differentiation by selecting a single price applied uniformly across all streets and durations. We refer to this as the linear pricing policy.

In order to write our objective function, we predefine the portion of each parking duration that falls under short-term and long-term pricing. Specifically, for each street i and duration-type ϕ , we compute the number of intervals that would be charged at the short-term rate (capped at four 15-minute intervals) and the remaining hours, if any, that would be charged at the long-term rate as: $D_{i,\phi,S} = \min(\mu_{i,\phi}, 4)$ and $D_{i,\phi,L} = \max(0, \mu_{i,\phi} - 4)$. Then, we aim to maximize daily revenue, subject to the problem constraints, as presented in the following mathematical formulation where Table EC.6 summarizes the notation used:

$$\begin{aligned} \max \quad & \sum_{t=1}^T \sum_{i=1}^N \sum_{j=1}^N \sum_{m \in M} \frac{1}{4} \left[p_j^S a_{ijtm}^{LS} \mu_{i,L} + \sum_{\phi \in \{S,L\}} p_j^\phi (a_{ijtm}^{SS} D_{i,S,\phi} + \sum_{v \in \{S,L\}} a_{ijtm}^{vL} D_{i,L,\phi}) \right] \\ \text{s.t.} \quad & \text{Constraints (1)–(9),} \\ & \text{Non-negativity constraints} \end{aligned}$$

The above formulation constitutes a nonconvex mixed-integer nonlinear program (MINLP) due to the nonlinear and fractional nature of the expressions defining ω_{ij} and $\rho_{v\phi}^{ijm}$, both of which are functions of the decision variable p_j^τ . To reflect real-world pricing practices and reduce model complexity, we assume that prices are selected from a finite set of discrete options, as municipalities typically define parking fees using discrete levels rather than continuous values. Let $\alpha_b \in \mathcal{P}$ denote the b -th price couple with two prices each, and let γ_{bj} be a binary decision variable indicating whether price couple α_b is selected for street j . Then, the selected price for each street is determined by the following constraints:

$$p_j^\tau = \sum_{b \in PL} \alpha_b^\tau \gamma_{bj}, \quad \forall j \in N, \quad (10)$$

$$\sum_{b \in PL} \gamma_{bj} = 1, \quad \forall j \in N, \quad (11)$$

$$\gamma_{bj} \in \{0, 1\}. \quad \forall b \in PL, \forall j \in N. \quad (12)$$

This discretization allows us to precompute all feasible values of ω and ρ for each (i, j, t, m) combination and price couple b . The resulting values, $\bar{\omega}_{ijb}$ and $\bar{\rho}_{v\phi}^{ijmb}$, are then fixed as parameters.

Consequently, both $\bar{N}_{itb}$ and $\bar{y}_{ijtm}^{v\phi}$ can also be computed in advance and treated as parameters in the model. As a result, the parameter $\bar{y}_{ijtm}^{v\phi}$ is defined as:

$$\bar{y}_{ijtm}^{v\phi} = \begin{cases} \delta_{mi} \bar{N}_{itb} \bar{\omega}_{ijb} \bar{\rho}_{v\phi}^{ijm} \kappa_i, & \text{if } v = S, \phi = S, L, \\ \delta_{mi} \bar{N}_{itb} \bar{\omega}_{ijb} \bar{\rho}_{v\phi}^{ijm} (1 - \kappa_i), & \text{if } v = L, \phi = S, L. \end{cases} \quad (13)$$

Furthermore, our original objective function includes multiplication of pricing variables p_j^S and p_j^L with the allocation variables $a_{ijtm}^{v\phi}$. As both p and a are continuous decision variables, this introduces bilinearity which results in computationally intractable model for large instances. To linearize this expression, we again leverage the price discretization, and define auxiliary continuous variables $\theta_{ijtm}^{v\phi}$ to represent the product $a_{ijtm}^{v\phi} \gamma_{bj}$. Finally, we add the following linear constraints to maintain logical consistency:

$$a_{ijtm}^{v\phi} - \theta_{ijtm}^{v\phi} \leq Cap_j (1 - \gamma_{bj}), \quad (14)$$

$$0 \leq \theta_{ijtm}^{v\phi} \leq a_{ijtm}^{v\phi}, \quad (15)$$

$$\theta_{ijtm}^{v\phi} \leq Cap_j \gamma_{bj}. \quad (16)$$

As a result, with the discretization of pricing decision, our model turns into a mixed-integer linear programming as shown in Section EC.2.1. [While prescriptive models with endogenous demand may raise optimism concerns, our formulation mitigates this by governing demand shifts through discrete price levels and fixed utility coefficients estimated from stated choice data. Consequently, behavioral responses are bounded by empirically grounded substitution patterns, which limits the potential for unrealistic optimism.]

5. Computational Studies

This section presents the computational studies conducted to evaluate the proposed pricing models. We begin by describing our real-life transaction data and explaining the estimation of key model parameters derived from it. Next, we outline the data selection steps used to decide on the streets and their neighbors to be included in the optimization model. Finally, we report the results of our comparative analysis across pricing strategies, and highlight the impact of different pricing structures on revenue and occupancy outcomes.

5.1. Data Description and Estimation of Problem Parameters

Our analysis draws on a rich dataset provided by our partner organization, which manages public parking facilities in one of the most densely populated cities in North America. In this city, drivers are required to prepay for parking rather than paying for the exact time their vehicle remains in the spot. Parking payments can be made either at street parking meters located near the spaces or through mobile parking applications. Our dataset contains over 5.7 million individual parking transactions across more than a thousand street segments from April 2019 to December 2020. For each transaction, we observe the exact time of payment, location, and payment method (cash, credit, mobile), along with parking price and payment amount. Figure EC.6 illustrates monthly parking transaction volumes alongside the share of mobile payments, which highlights temporal variation in demand and the steady increase in mobile adoption over time. In Section EC.8, Figure EC.7 presents a heatmap of adjusted parking transactions across street intersections, where transaction counts are normalized by local capacity to reflect relative demand. The map shows that parking demand varies significantly across streets, with certain intersections experiencing disproportionately high usage relative to their capacity.

We begin by constructing the key variables used in our modeling framework using the transaction data. To do so, all payment durations are discretized into 15-minute intervals prior to analysis to simplify computational tasks in the mathematical modeling and estimation processes. To estimate localized demand conditions, we construct a distance-weighted measure of parking activity, denoted N_{it} , which reflects the spatial density of transactions around street i at time interval t . For each unique combination of street and time interval, we count the number of transactions and compute N_{it} as a weighted sum of transaction counts on all streets within the same time interval, with weights inversely proportional to their geographic distance from street i . Specifically, the demand measure is given by:

$$N_{it} = \sum_{j \in \mathcal{I}_t} \frac{1}{\max(d_{ij}, \varepsilon)} \text{Transactions}_{jt},$$

where $\mathcal{I}_t$ is the set of all streets with at least one transaction at time t , d_{ij} denote the distance between street i and another street j , and ε is a small constant to avoid division by zero. This formulation ensures that nearby streets contribute more heavily to N_{it} to capture the notion that parking conditions are spatially interdependent.

As described in the modeling part, we operationalize the duration structure using the two representative duration values, μ_{iS} and μ_{iL} , and the proportion of short-duration drivers for each street,

κ_i . To estimate these quantities, we use a Gaussian Mixture Model (GMM) approach applied to the distribution of discretized payment durations, details can be found in Section EC.2.2. In addition to duration estimates, we calculate the share of mobile payment usage on each street, δ_{mi} . This behavioral measure helps us calibrate differences in drivers' flexibility to extend their stay. As previously discussed, drivers using cash or card must return to the meter to extend their session, and they incur a hassle cost that is embedded in the regret term via h_m . In contrast, mobile users can extend their payment remotely, which reduces their hassle and changes the perceived risk of underpaying initially. For each street, mobile payment share is computed as the proportion of transactions paid via mobile applications, based on the recorded payment method.

Lastly, to ensure comparability between regrets coming from monetary overpayment and inconvenience from extending, we express the hassle cost h_m as a fraction of the regret incurred from overpaying for a long duration. Specifically, we define $h_m = \Psi_m p_f^L(\mu_{iL} - \mu_{iS})$, where $\Psi_m \in [0, 1]$ captures the relative inconvenience of extending a parking session under payment method m . This normalization allows both regret terms to be represented in monetary units and ensures that duration choice probabilities reflect meaningful trade-offs between financial and convenience-based considerations. We utilize lower values of Ψ_m correspond to mobile payment because of its convenience, while higher values to reflect greater inconvenience of other payment methods.

5.2. Neighborhood Selection Procedure

To ensure data reliability for our optimization model, we apply two filtering criteria to eliminate streets with insufficient activity. First, we exclude streets where the number of recorded observations per parking spot per day was less than three. This threshold reflects the structure of our data, which includes 36 intervals of 15 minutes each day, and the fact that the maximum allowed parking duration is three hours. That means, fewer than three observations suggest limited turnover which makes demand estimation unreliable. Second, we exclude streets that had no new arrivals during at least half of the daily intervals which indicates low usage patterns that may not support meaningful pricing adjustments.

After applying these filters, we group streets based on their physical proximity using the algorithm given in Section EC.2.3. The algorithm systematically constructs spatially clustered and computationally tractable street groups for use in neighborhood-level parking optimization. It begins by defining key parameters: the maximum number of expansion steps (MAX_ITER), a distance threshold (DIST_THRESHOLD), and allowable group size bounds (MIN_GROUP_SIZE and MAX_GROUP_SIZE). The procedure iterates through each street in the preprocessed dataset and

initiates a bounded neighborhood expansion for streets not yet assigned to a group. Starting from a single street, the algorithm identifies all neighboring streets within a `DIST_THRESHOLD`-meter radius, then recursively includes their neighbors in subsequent iterations, while excluding any streets already processed. This expansion continues for up to `MAX_ITER` iterations or until no new streets are found. If the resulting group contains fewer than 6 or more than 8 streets, it is discarded to balance model tractability, behavioral diversity, and interpretability. Valid groups are finalized by marking all included streets as processed to ensure disjoint group assignment. As a result of this procedure, we obtained 11 groups with varying number of neighboring streets including 73 streets in total. These groups differ in total parking capacity, the distribution of estimated duration parameters, and mobile application usage patterns, which reflects local heterogeneity in parking behavior. Descriptive statistics for these groups are presented in Table EC.8.

5.3. Computational Results

To evaluate the performance of different pricing strategies, our computational studies are conducted on a MacBook Pro M1 chip computer, where we allow models to stop after achieving 1% optimality gap. We run our pricing models using predefined short- and long-term price pairs defined as follows:

Table 4 Feasible Progressive Pricing Combinations

Short-term Price	Long-term Price(s)
3	3, 4, 5
4	4, 5, 6
5	5, 6, 7
6	6, 7
7	7

This set includes both constant pricing such as setting the same price for both short- and long-term pricing, and progressive options where the long-term price exceeds the short-term price. We focus only on non-decreasing form of progressive pricing as it is the commonly used structure by cities around the world (Ornelas et al. 2023). Combinations such as (3,6), (3,7), or (4,7) are excluded to avoid overly steep pricing differentials. The selected pairs aim to balance behavioral influence with operational usage possibility. To evaluate the performance of our model, we conduct two complementary sensitivity analyses: one controlled, and one comparative across all groups. For each scenario, we run our mathematical model three times for each group, using data from three distinct days. This variation allows us to account for day-to-day fluctuations in baseline parking demand. Additionally, to ensure that our findings are not driven by the randomness embedded in the MNL model, we complete three independent runs for each setting, each with a different set

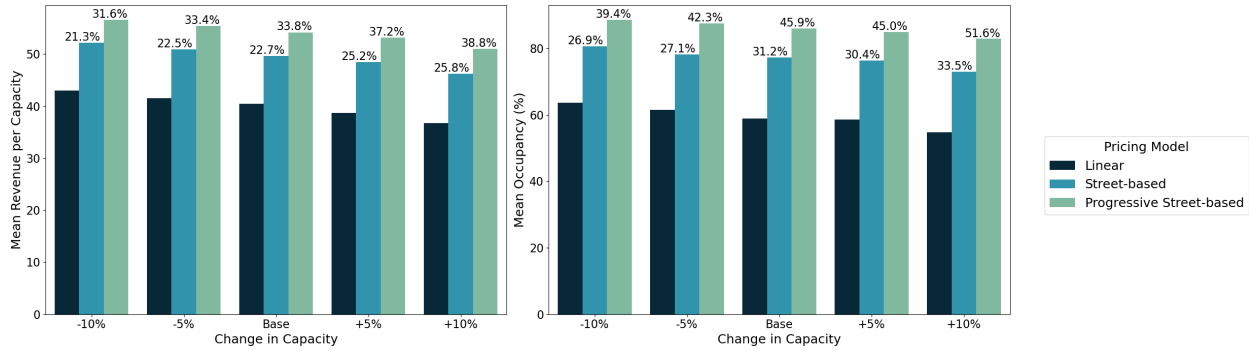
of randomly drawn error terms. These sensitivity analyses are motivated by practical variability observed in real-life parking environments. Municipalities regularly encounter fluctuations in user composition, infrastructure constraints, and behavioral patterns. By evaluating how our proposed strategies perform under different plausible scenarios, we aim to provide a more practice-driven foundation for policy decisions and support city planners in adapting pricing structures to their specific operating conditions.

5.3.1. Controlled Sensitivity Analysis We begin by selecting one representative group at random and examine how changes in system parameters affect pricing outcomes. Specifically, we systematically change capacity, proportion of user types, and average parking durations, and evaluate their effects on occupancy and revenue under three pricing strategies. For parameters other than duration, we modify the base levels by $\pm 5\%$ and $\pm 10\%$; for average durations, we limit the change to $\pm 10\%$ since $\pm 5\%$ shifts do not meaningfully alter the 15-minute interval coverage used in the pricing policy.

Change in the Capacity. To assess the effect of changes in the parking capacity, we run our model under multiple capacity scenarios for three pricing models. For each scenario, we computed average revenue per capacity and average occupancy by aggregating outputs across all streets in a group. As shown in Figure 2, as capacity increases, both revenue per capacity and occupancy decrease under all pricing models. This emphasize the supply and demand mismatch. When more spaces are available but demand do not change significantly, average revenue per unit capacity declines due to underutilization. However, as shown in the figures, progressive street-based pricing yields higher occupancy and greater revenue per space under all scenarios, and its advantage becomes more pronounced with added capacity. This is due to its ability to discourage long stays and promote turnover, thus making better use of surplus spaces. On the other hand, linear street-based pricing offers moderate improvements but lacks the duration-based incentives necessary to capitalize on increased supply.

Change in the Proportion of Short-term Parkers. As κ_i , the proportion of short-term parkers, increases, both revenue per capacity and occupancy decrease across all pricing models. This result arises because short-term parkers contribute less total revenue and occupy space for a shorter time, which, without sufficient turnover volume, leads to lower average utilization and income per unit of capacity. While short stays increase the potential for turnover, the actual benefit depends on whether that turnover is realized at sufficient scale. Without a proportional increase in arrival rate, replacing

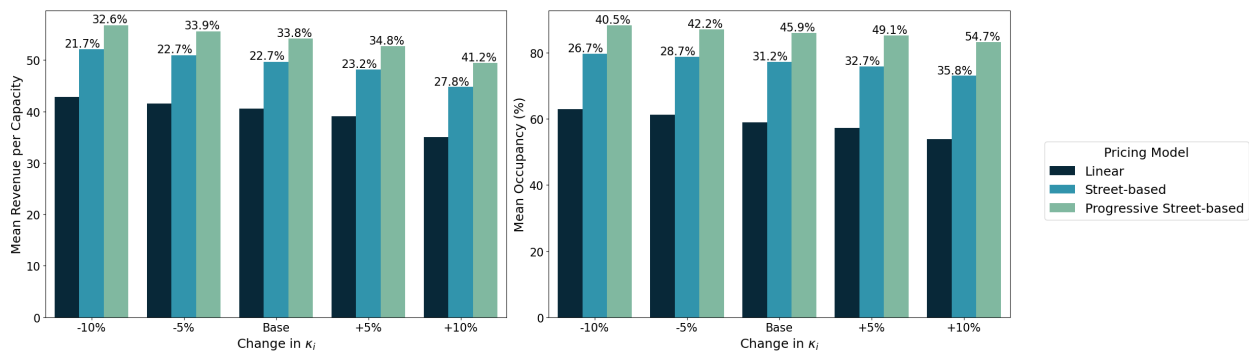
Figure 2 Impact of Pricing Policies Under Changing Capacity



long-stay drivers with short-stay drivers results in more idle time per spot and thus reduced overall system efficiency.

Our analysis shows that progressive street-based pricing consistently outperforms the other models, with its relative advantage growing steadily as κ_i increases (with minor deviations possible due to the 1% optimality gap allowed in model solutions). For instance, revenue per capacity under progressive street-based pricing rises from 32.6% to 41.2% above linear pricing, while occupancy gains grow from 40.5% to 54.7%. These results indicate that progressive street-based pricing becomes more advantageous as the proportion of short-term users grows. Linear pricing, by contrast, applies the same rate to all durations and fails to take advantage of this behavioral shift. Thus, even though overall revenue and occupancy fall due to shorter average usage, progressive street-based pricing captures more value from the same demand structure and becomes comparatively more effective as κ_i increases.

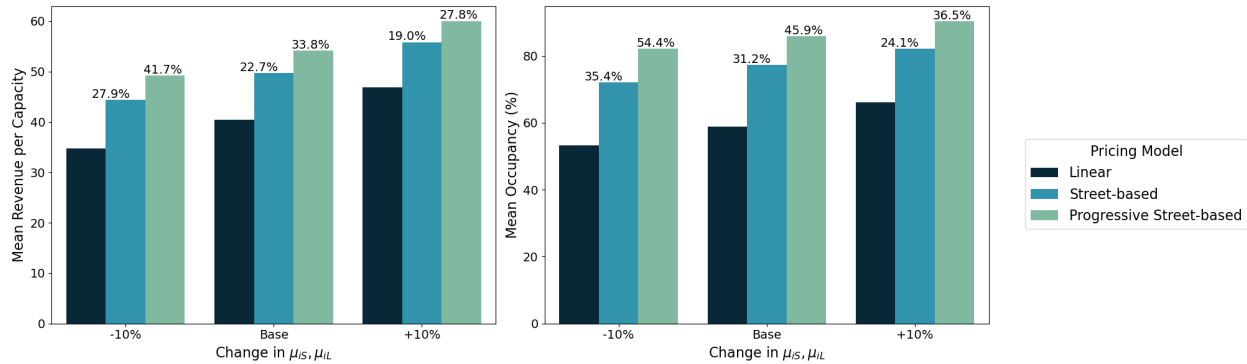
Figure 3 Impact of Pricing Policies Under Changing Proportion of Short-term Parkers



Change in the Mean Short- and Long-term Parking Duration. We also change the short- and long-term parking durations by $\pm 10\%$. Figure 4 shows that as the average parking duration increases, both occupancy and revenue per capacity increase across all pricing models. Longer stays translate

into greater cumulative use of parking space and more revenue collected per transaction, especially when demand levels remain constant. However, the relative benefit of progressive street-based pricing declines. When mean durations are short, progressive street-based pricing leads to much higher occupancy and revenue compared to linear pricing. As durations increase, all users stay long regardless of price structure, so the difference between pricing models narrows. This result is consistent with our earlier findings. Previously, we observed that when the system shifts toward long-term parking behavior, progressive street-based pricing loses its relative advantage. Although the increase in duration comes from a direct change in the duration parameter rather than a behavioral shift, the outcome is similar: when most users stay for a long time, progressive street-based pricing has less to work with. Its main advantage is encouraging quick turnover by charging more for longer stays, but when nearly everyone parks for a long time anyway, that incentive loses its impact.

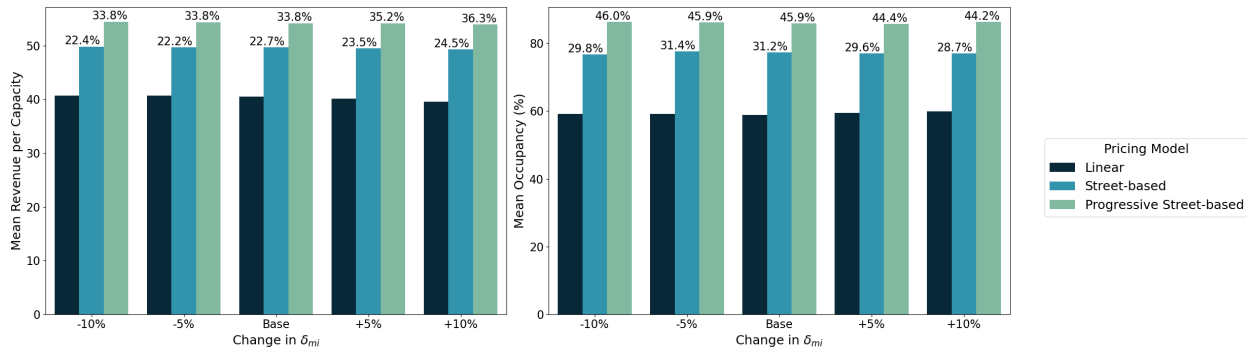
Figure 4 Impact of Pricing Policies Under Changing Parking Durations



Change in the Proportion of Mobile Payers. Finally, we test the effect of proportion of short-term parkers using the proportion of mobile payers, δ_{mi} . Mobile payment users face lower friction and are more inclined to select short-term parking due to reduced effort in transaction and potential for real-time extensions. With the increase in the proportion of mobile payers, both occupancy and revenue per capacity remain relatively stable across all pricing models. While progressive street-based pricing continues to yield higher occupancy than linear pricing across all levels, its relative advantage narrows slightly as the proportion of mobile payers increases. This outcome reflects the system reaching a saturation point in space utilization: as more users shift to short-term parking due to mobile payment convenience, a greater share of demand is already aligned with high-turnover behavior, even under less sophisticated pricing models. Consequently, the marginal occupancy gain achieved by progressive street-based pricing diminishes, since fewer users attempt long stays to

begin with. However, while occupancy saturates, revenue per capacity grows with turnover, and progressive street-based pricing continues extracting value efficiently.

Figure 5 Impact of Pricing Policies Under Varying Mobile Payment Adoption



5.3.2. Combined Group-Level Sensitivity Analysis To generalize beyond a single group, we extend our analysis to the remaining groups by aggregating results across shared street characteristics. After running the model in total 27 times (3 pricing model $\times$ 3 days $\times$ 3 randomization), we classify all groups by quantiles of street capacity, dominant user type proportions, or average parking durations, and compare average outcomes across pricing strategies within these classes. For each class, Table 5 reports the mean revenue per capacity and mean occupancy achieved under linear pricing, alongside the percentage improvements observed under street-based and progressive street-based pricing models.

Table 5 Combined Group-Level Sensitivity Analysis Results

Metric	Class	Mean Revenue Per Capacity (\$)			Mean Occupancy (%)		
		% Δ from Linear Pricing			% Δ from Linear Pricing		
		Linear	Street-based	Progressive Street-based	Linear	Street-based	Progressive Street-based
Capacity	Low	55.1	10.4%	13.2%	80.2	15.1%	17.9%
	Medium	36.0	35.5%	48.5%	53.1	56.7%	60.3%
	High	31.3	15.9%	26.9%	46.3	33.0%	42.4%
Duration	Short	42.1	17.1%	24.1%	62.4	28.2%	35.0%
	Medium	39.9	21.8%	32.0%	58.2	38.4%	41.9%
	Long	46.3	14.6%	18.7%	67.4	21.7%	24.6%
Mobile	Low	48.27	12.7%	17.8%	70.00	21.3%	25.1%
	Medium	40.22	17.9%	27.3%	59.88	30.7%	37.6%
	High	39.53	23.6%	30.6%	57.81	37.0%	39.7%

Grouping based on Capacity. To better understand pricing performance across heterogeneous street groups, we used all our street groups and classified streets into capacity-based clusters based on the quantile of available space per street as shown in Table EC.9. Table 5 presents average revenue per capacity and occupancy for each class under different pricing models. Linear pricing

shows diminishing returns as capacity increases, as it fails to match demand variability and leading to lower space utilization; therefore, it is inefficient in capturing revenue in higher-capacity areas. Progressive street-based pricing, by contrast, performs best across all capacity groups, with its strongest impact seen in medium-capacity streets. These areas are typically the most balanced in terms of supply and demand, and allow pricing mechanisms to influence behavior effectively. In low-capacity areas, demand tends to exceed supply, so occupancy remains high regardless of the pricing model. As a result, the room for improvement is limited, and the gains from progressive street-based pricing are modest. On the other hand, high-capacity regions likely have excess supply relative to demand. Even with optimized pricing, unutilized spaces lead to lower occupancy and, consequently, lower revenue per unit of capacity.

Together, these findings indicate that capacity adjustments remain effective when paired with duration-sensitive pricing models. However, the standalone impact of capacity changes diminishes without a pricing structure that activates turnover. Therefore, pricing design plays a critical role in extracting value from available supply.

Grouping based on Mean Payment Duration. For a broader analysis which uses all street groups, we consider mean duration and proportion of short term parkers parameters together. To capture their joint influence on expected parking duration, we use a combined measure:

$$\mu_i^{combined} = \kappa_i \mu_{iS} + (1 - \kappa_i) \mu_{iL}.$$

Then, we use $\mu_i^{combined}$ to classify streets into short-, medium-, and long-duration groups. As shown in Table 5, progressive street-based pricing performs consistently well across different parking duration profiles. However, as in the capacity analysis, its performance exhibits a sweet spot where the highest gains in both revenue per capacity and occupancy are observed in the medium-duration group. These streets offer sufficient turnover for dynamic pricing to segment users effectively, while also allowing enough parking time to extract differentiated payments. They strike a balance between space availability and behavioral responsiveness. In contrast, long-duration streets yield more modest improvements, as users tend to occupy spots for extended periods regardless of price. This limits both turnover and responsiveness, and reduces the effectiveness of progressive street-based pricing structures in boosting revenue and space utilization.

Grouping based on Proportion of Mobile Payers. Finally, we check the change in the revenue per capacity and occupancy using the grouping based on the mobile payment proportion. As mobile payment adoption increases, both revenue and occupancy improve with higher mobile payment

share, and the relative advantage of progressive street-based pricing increases. This suggests that when mobile payment becomes dominant behavioral alignment emerge, and it enables progressive street-based pricing to function as intended by segmenting users effectively and maximizing both revenue and occupancy.

Overall, these findings provide several operational recommendations. First, as shown in the capacity analysis, simply increasing parking supply does not effectively address urban parking challenges. Without complementary pricing interventions, added capacity often leads to underutilization and reduced efficiency. Cities need pricing strategies that actively shape demand and improve how existing supply is used, rather than relying on expansion alone. Second, municipalities currently relying on linear pricing are likely leaving substantial revenue on the table. Linear pricing fails to adapt to demand variability, duration heterogeneity, or supply expansion. It should be limited to areas with low demand and minimal turnover needs. In environments where full implementation of time-sensitive pricing is not yet feasible, linear street-based pricing offers a practical intermediate step. It still provides moderate revenue and occupancy improvements through spatial differentiation while requiring less infrastructure than time-based policies. However, to unlock the full potential of pricing interventions, cities should transition toward time-based progressive street-based pricing.

Our analysis combining both controlled simulations and observational group-level comparisons shows that the effectiveness of progressive street-based pricing is influenced by both street-level characteristics and the behavioral profiles of users. In the controlled setting, where individual parameters are varied while others are held constant, progressive street-based pricing consistently outperforms alternatives, especially as parking capacity increases or when the proportion of short-term or mobile users rises. Furthermore, when applied across heterogeneous street clusters, the model's relative advantage peaks in medium-duration and medium-capacity environments, where behavioral responsiveness and supply conditions are optimally aligned. Together, they show that progressive street-based pricing is most effective where demand is strong, behavior is flexible, and space is constrained, such as dense commercial zones with high mobile payment adoption. In contrast, linear street-based pricing offers only moderate improvements and may be suitable in stable, low-turnover contexts where time-based differentiation is impractical.

6. Conclusion

This study examined how parking attributes influence drivers' substitution behavior and incorporates these insights into a pricing optimization model. We first designed an online survey to capture

how participants' utility changes in response to variations in parking attribute levels. The findings suggested that the utility of participants linearly decrease with the increase in the *Parking Price*, while the decrease is nonlinear when we move towards the unwanted levels of *Distance to Destination* and *Likelihood of Extension*. Post hoc analyses further revealed that participants in the mobile application condition place greater emphasis on parking price compared to those in the cash and credit card conditions. This indeed shows that convenience significantly affects choice behaviors.

Furthermore, to operationalize these insights, we proposed a mathematical model that determines optimal parking prices while endogenously accounting for both demand elasticity and attribute-driven substitution behavior. Each driver was modeled as making a probabilistic choice over available parking options based on the attributes which also touch on the behavioral realism that is often absent in the prior models. The model integrated these choice probabilities into a street-level pricing framework, which allows demand to shift not only across locations but also across duration types, namely, short- and long-term parking. Price decisions were optimized under capacity constraints, and alternative pricing structures were evaluated, including linear pricing, and linear and progressive street-based pricing. The overarching conclusion is that municipalities can benefit substantially from adopting differentiated pricing schemes. Specifically, the results showed that even simple spatial price differentiation yields notable revenue improvements, while progressive street-based pricing further enhances both revenue and occupancy.

Our results provide actionable managerial insights for municipalities aiming to optimize parking management through targeted policies. First, our empirical findings highlight the importance of integrating behavioral attributes beyond price and location into parking policy design. In particular, we introduce the *Likelihood of Extension* as a novel attribute, and show that it significantly influences parking choices. This suggests that municipalities can enhance drivers' parking experience by increasing perceived flexibility through technology-enabled extensions. Second, our mathematical model provides a scalable framework for setting duration and street-based prices while accounting for substitution across both location and parking duration. Unlike prior models, which treat demand as static and homogeneous, our model incorporates empirically estimated choice probabilities and allows drivers to adjust their decisions based on price, distance, and extension likelihood. This enables more realistic demand allocation and better pricing outcomes. Finally, we show that adopting linear or progressive street-based pricing schemes yields clear advantages over linear pricing. Linear street-based pricing alone leads to substantial revenue gains by aligning prices with localized demand patterns. When paired with progressive pricing, the model further improves both occupancy

and revenue. Our analysis shows that the effectiveness of strategic pricing policies is influenced by both street-level characteristics and the behavioral profiles of users. The relative advantage of having a differentiated pricing policy compared to linear pricing peaks in medium duration and medium capacity environments where behavioral responsiveness and supply conditions are optimally aligned. These findings support a move away from one-size-fits-all pricing policies toward behaviorally grounded, spatially adaptive strategies.

Despite its contributions, this study has several limitations. First, the behavioral component relies on stated preference data from a choice-based conjoint survey, which may not fully capture actual behavior under real-world constraints or stress conditions. Although we carefully designed the survey and analyzed the collected data, hypothetical bias and simplification of the decision context remain concerns. Second, the optimization model assumes relatively stable parking supply and demand conditions over the decision-making horizon. In practice, parking availability is subject to abrupt fluctuations due to road closures, special events, or construction work. While the model does not account for these short-term disruptions, our extensive sensitivity analysis mitigates this limitation to the extent possible. Third, the discretization of price and precomputation of behavioral parameters improve tractability but limit the granularity and flexibility of pricing levels, potentially constraining optimality in edge cases. Finally, while the model incorporates substitution across streets and between short and long durations, it does not consider how parking choices interact with other modes of transportation, which could affect demand patterns and limit the generalizability of the pricing strategy in broader urban mobility contexts.

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Appendix

EC.1. Survey Design Complementary

To understand how individuals make parking decisions when presented with alternatives that differ across key attributes, we designed an online choice-based survey that captures participants' preferences under realistic parking scenarios. In Section EC.1.1, we describe the overall structure of the survey, including participant selection and the instructional materials provided before the choice tasks. In Section EC.1.2, we detail the comprehension and attention check questions designed to ensure participants understood the task and the attributes involved before moving to the choice tasks. Section EC.1.3 explains the construction of the choice tasks and presents a sample question to illustrate the decision context participants encountered. Section EC.1.4 outlines the final part of the survey, where we collected demographic information, and asked about participants' typical parking behavior and payment preferences. In Section EC.1.5 and beyond, we provide further details of the data analysis, including our STAN model, as well as a discussion of estimation accuracy and the robustness of the results.

EC.1.1. Survey Instructions

Using an online survey, we examine how different parking attributes influence drivers' parking decisions. To mitigate selection bias, participants were recruited through the Prolific platform. Eligibility was restricted to individuals aged 18 and older, aligning with the legal driving age and the requirement to hold a valid driver's license. Moreover, recruitment focused on English-speaking countries such as the United States and Canada to eliminate the need for survey translation. Additional participant information is provided in the subsection covering the end-of-survey questions.

Participants were randomly assigned to one of three experimental conditions, each corresponding to a different payment method: cash, credit card, or mobile application. In our survey design, we deliberately chose not to include payment method as one of the attributes varied within the choice tasks. This design decision was driven by both technical and methodological considerations. Technically, incorporating payment method as an attribute within the conjoint design would have required participants to evaluate and compare different payment mechanisms simultaneously across alternatives within a single task. However, this setup would not reflect the actual decision process in real-life on-street parking, where drivers may choose their preferred payment method at the time of payment, but this choice does not influence which parking spot they select, as all nearby spots typically support the same set of payment options. From a methodological perspective,

implementing payment method as a between-subject factor avoids biases associated with within-subject comparisons. Presenting multiple payment types to the same participant may draw their attention to the experimental manipulation and change their natural decision-making process. This approach aligns with experimental best practices and is supported in the literature as a strategy to reduce behavioral biases in economic experiments (Charness et al. 2012).

The study began with detailed instructions about the survey. Participants were informed that they would complete a total of 20 choice tasks, each displaying three different on-street parking alternatives alongside a “no-choice” option. The “no-choice” option was described as choosing to park in a garage at a substantially higher cost than any of the on-street alternatives. Before beginning the tasks, participants were also familiarized with the structure of the alternatives and the specific attributes that would vary across choice sets. Figure EC.1 presents an example of the instruction screen shown to participants.

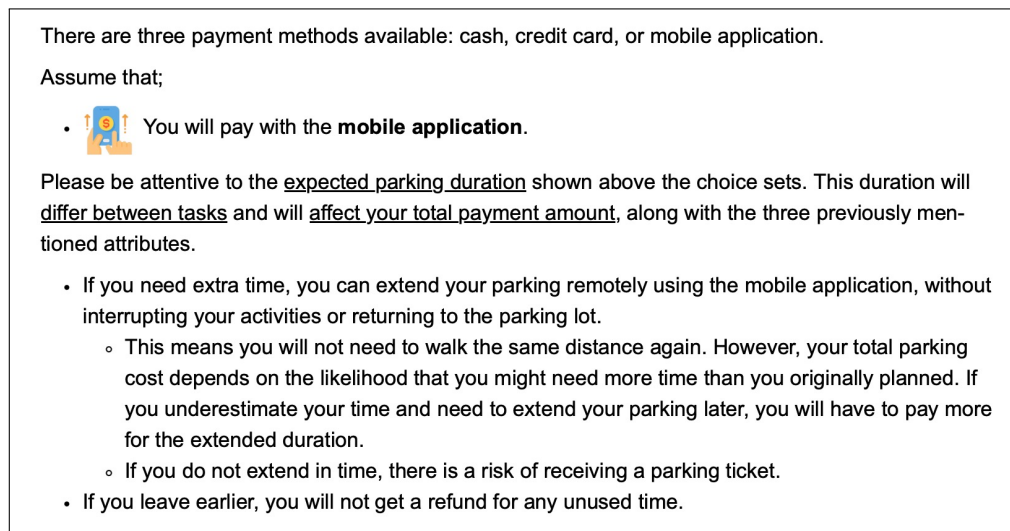
Figure EC.1 Instructions Available to Each Participant before Moving to Choice Tasks

Instructions		
You will now be presented with a total of 20 choice tasks. In every choice task, you will see three different parking alternatives. Each time, the "none" option should be interpreted as parking in a garage agreeing to pay much more than paid on-street parking. It is important that you approach every choice task as an independent one, focusing only on comparing the three parking alternatives in front of you. Select the most attractive alternative or the none option. Before we proceed, we wanted you to familiarize yourself with the type of alternatives you will be asked to choose from. For each choice task, imagine that the parking alternatives differ only between three attributes. The table below summarizes those attributes and the different levels that describe them.		
Attribute	Levels	Explanation
Parking Price	3 \$/hour 4 \$/hour 5 \$/hour 6 \$/hour 7 \$/hour	It represents the hourly cost of parking. The hourly parking price shown in each choice will be adjusted based on the time you enter into the payment station, and you will pay accordingly.
Distance to Destination	50 meters (164 feet) 100 meters (328 feet) 200 meters (656 feet) 400 meters (1312 feet) 800 meters (2624 feet)	The walking distance from the parking spot to the intended destination.
Likelihood of Extension	<20% 20%-40% 41%-60% 61%-80% >80%	It shows the approximate probability of needing to extend your parking duration.

Following this general introduction, depending on the randomly assigned condition, participants were provided with instructions describing how their assigned payment method would influence

their ability to modify the parking session. For example, those assigned to the mobile application condition were informed that they could extend their parking remotely without returning to the parking lot, whereas participants in the cash and credit card condition were told they would need to walk back to the kiosk to make any extension. Under all conditions, it was clarified that parking payments were not refundable if the participant left earlier than expected and that failure to extend the session on time might result in a parking ticket. Figure EC.2 presents an example of the instruction screen shown to participants assigned to the mobile payment condition.

Figure EC.2 Instructions regarding Payment Method Assignment before Moving to Choice Tasks



EC.1.2. Attention Check Questions

To ensure participants understood the task and the meaning of the attributes, we asked them five questions prior to beginning the choice tasks. Two of these were attention checks. The first question (Figure EC.3) asked participants to identify an attribute that was not part of the study, while the second one (Figure EC.4) asked them to recall their assigned payment method. We did not disqualify any participants for answering these questions incorrectly, as they can view their payment method and attributes while responding to the choice tasks. However, we conducted a robustness test where we discard these observations and repeat the analysis. Section EC.1.7 shows that our findings are robust.

The remaining three questions tested participants' understanding of the core concepts in the survey, including how parking payments would be calculated, what the likelihood of extension represented, and how the distance attribute should be interpreted. Participants had to answer these three conceptual questions correctly to proceed with the survey.

Figure EC.3 First Attention Check Question

**This is an attention check.*

Select the attribute that will NOT be considered in this survey:

❶ Choose one of the following answers

☐ Parking Price

☐ Likelihood of Extension

☐ Day of Week

☐ Distance to Destination

Figure EC.4 Second Attention Check Question

**This is an attention check.*

Which payment method have you been instructed to use?

❶ Choose one of the following answers

☐ Cash

☐ Credit Card

☐ Mobile Application

☐ All three alternatives

EC.1.3. Sample Survey Question

After participants completed the instructional and comprehension phases of the survey, they were redirected to the choice-based conjoint (CBC) module programmed in Sawtooth Software. In this module, each participant was shown a total of 20 choice tasks. We generated our choice tasks using the *%MktEx* and *%ChoiceEff* macros in SAS. First, we used the *%MktEx* macro to create a candidate set of alternatives to ensure a structured and balanced pool from which the final design could be drawn. Then, we applied the *%ChoiceEff* macro to construct an efficient experimental design by maximizing D-efficiency. While constructing the designs, we benefited from the restriction flexibility of the *%ChoiceEff* macro to eliminate dominated alternatives to ensure that choice sets remained realistic and aligned with rational decision-making principles. We created five different designs, each containing 20 tasks. A detailed explanation of all macros can be found in the book by Kuhfeld (2003). The literature also emphasizes the significance of keeping nearly-balanced designs, as imbalances may cause respondents to modify their answers based on perceived patterns or to react differently to levels presented more frequently (Kuhfeld 2003, pp. 253). In our case, the ideal scenario would involve a perfectly balanced distribution of attribute levels, with each level appearing exactly 12 times (calculated as $20 \text{ tasks} \times 3 \text{ alternatives} \div 5 \text{ levels} = 12$). However, our efficient design introduced slight deviations, with some levels appearing only 11 times. Despite these slight frequency variations in the final design, it is still balanced enough to mitigate risks. To control for any design-related effects across experimental conditions, we also ensured that the same sequence of design versions was used across all payment method groups (respondent 1 of each payment method saw the same version, as did respondent 2, and so on).

In addition to the randomized attribute combinations, the survey incorporated varying expected parking durations, which were displayed to participants at the top of the screen during the CBC tasks as shown in Figure EC.5. There were five different expected parking duration ranges used in the

study: 5–20 minutes, 21–45 minutes, 46–70 minutes, 71–95 minutes, and 96–120 minutes. These durations were not randomly assigned but were systematically rotated using a Latin square design across blocks of four choice tasks. Each participant thus saw all five duration ranges exactly four times, with the order of presentation balanced across participants. This structure was implemented to minimize biases from question spillover (Bradley 1958).

Figure EC.5 Instructions regarding Payment Method Assignment before Moving to Choice Tasks

Assume your expected parking duration is **between 96 to 120 minutes**. The hourly parking price shown in each choice will be adjusted based on the time you enter into the payment station, and you will pay accordingly.

Remember that you will pay with **mobile application**.

- If you need extra time, you can extend your parking remotely using the mobile application, without interrupting your activities or returning to the parking lot.

This means you will not need to walk the same distance again. However, your total parking cost depends on the likelihood that you might need more time than you originally planned. If you underestimate your time and need to extend your parking later, you will have to pay more for the extended duration.

If you do not extend in time, there is a risk of receiving a parking ticket.

- If you leave earlier, you will not get a refund for any unused time.

If these were the only scenarios, which scenario would be preferable to you?

(2 of 20)

	4 \$/hour	5 \$/hour	7 \$/hour
Parking Price	4 \$/hour	5 \$/hour	7 \$/hour
Distance to Destination	400m (1312 feet)	50m (164 feet)	200m (656 feet)
Likelihood of Extension	61%-80%	20%-40%	<20%
	Select	Select	Select

NONE: I prefer using a garage, agreeing to pay a fixed daily rate of \$21 instead of an hourly fee.

EC.1.4. End of Survey Questionnaire

Upon completion of the choice tasks, participants were requested to respond several questions regarding to their demographic characteristics and spending habits. Table EC.1 lists the end of survey questions.

Table EC.1 End of Survey Questionnaire

<i>Variable</i>	<i>Question</i>	<i>Response scale</i>
Importance	The importance of attributes for my selection of parking was	Order attributes
Availability	Is there paid on-street parking in your area?	Yes, No, I do not know
Recency	How recently did you use paid on-street parking?	Within the last week, within the last month, within the last 3 months, More than 3 months ago, Never used it
Frequency	How frequently do you use paid on-street parking?	Once a week or less, 2-3 times a week, 4-5 times a week, More than 5 times a week
Monetary	How much do you spend on paid on-street parking per month?	\$9.99 or less, \$10-\$19.99, \$20-\$29.99, \$30-\$39.99, \$40-\$49.99, More than \$50
Parking Area	Where do you most frequently park when using paid on-street parking?	Residential area, commercial/business district, entertainment/recreation areas, public transportation hubs, other, prefer not to answer (PNA)
Usage	How likely are you to prefer paid on-street parking over other parking options in the future?	Very unlikely, unlikely, neutral, likely, very likely, PNA
Commute	How do you most frequently commute to your workplace/school?	Car, bus, walk, subway, train, bike, not applicable, other, PNA
Commute Frequency	How often do you commute to your workplace/school in a typical week?	Less than once a week, once, 2-3 times, 4+ times, not applicable, PNA
Transportation	What is your primary mode of transportation during a typical week?	Personal vehicle, bus, walk, cab, subway, train, other, PNA
Preference	Which payment method would you often prefer when you pay for your parking?	Order payment methods
Convenience	How convenient would it feel to pay with the option assigned to you?	Very inconvenient, inconvenient, neither inconvenient nor convenient, convenient, very convenient, PNA
Pain of Payment	How painful would it feel to pay with the option assigned to you?	Very painful, painful, neither painful nor pleasurable, pleasurable, very pleasurable, PNA
Mobile Usage	How likely are you to use mobile applications for street parking payment in the future?	Very unlikely, unlikely, neutral, likely, very likely, PNA
Area	Which best describes the area where you live?	Urban area, suburban area, rural area, other, PNA
Population	What is the approximate population of your area?	Less than 10,000 residents, 10,000 - 100,000 residents, 100,001 - 500,000 residents, 500,001 - 1,000,000 residents, More than 1,000,000 residents, PNA
Income	What is your annual household income?	Less than \$25,000, \$25,000-\$49,000, \$50,000-\$99,000, \$100,000-\$199,000, More than \$200,000, PNA
Adult	How many adults, including yourself, live in your household?	1, 2, 3, 4, Over 4, PNA
Phone	How many smartphones do you have?	None, 1, 2, 3, 4, Over 4, PNA
Cars	How many cars do you have in your household?	None, 1, 2, 3, Over 3, PNA
EVs	How many electric vehicles do you have in your household?	None, 1, 2, 3, Over 3, PNA

We presented summary statistics of the covariates used in the analysis in Table EC.2. This descriptive analysis of the end of survey questions revealed the following:

1. A total of 450 participants completed the survey. In terms of perceived importance, parking price was most frequently ranked as the most important attribute (49.3%), followed by distance to destination and likelihood of extension.
2. Regarding usage patterns, the majority had used on-street parking within the past month, most often once a week or less (65.3%). Reported spending was typically modest, with 41.8% spending less than \$10 per use.
3. Respondents were primarily from urban (47.1%) and suburban (43.1%) areas, with a wide distribution across income and age groups. The largest age group was 30–39 years (32.7%), and the majority reported annual household incomes between \$50,000 and \$199,000.

Table EC.2 Survey Statistics (N = 450)

Importance 1	Importance 2	Importance 3
Distance to Destination 28.0% (126)	Distance to Destination 40.0% (180)	Distance to Destination 32.0% (144)
Likelihood of Extension 22.7% (102)	Likelihood of Extension 28.0% (126)	Likelihood of Extension 49.3% (222)
Parking Price 49.3% (222)	Parking Price 32.0% (144)	Parking Price 18.7% (84)
Recency of Use	Usage Frequency	Monetary Spend
Within last week 31.6% (142)	Once/week or less 65.3% (294)	\$9.99 or less 41.8% (188)
Within last month 28.4% (128)	2–3 times/week 23.6% (106)	\$10–19.99 24.9% (112)
Within 3 months 13.6% (61)	4–5 times/week 4.0% (18)	\$20–29.99 12.7% (57)
3 months ago 20.7% (93)	5 times/week 1.3% (6)	\$30–39.99 6.2% (28)
Never used 5.8% (26)	Never used 5.8% (26)	\$40–49.99 3.6% (16)
		\$50 5.1% (23)
		Never used 5.8% (26)
Area of Residence	Income	Age
Rural 9.1% (41)	< \$25,000 5.1% (23)	18–19 0.4% (2)
Suburban 43.1% (194)	\$25k–49k 12.7% (57)	20–29 22.4% (101)
Urban 47.1% (212)	\$50k–99k 39.1% (176)	30–39 32.7% (147)
Other 0.4% (2)	\$100k–199k 34.9% (157)	40–49 19.6% (88)
Prefer not to say 0.2% (1)	\$200k 6.9% (31)	50–59 15.1% (68)
	Prefer not to say 1.3% (6)	60–69 8.4% (38)
		70+ 1.3% (6)

EC.1.5. STAN Model

The hierarchical model implemented in Python is written as:

$$y_{ijk} \sim \text{CategoricalLogit} \left(\sum_{a=1}^A \sum_{l=1}^{L_a-1} \beta_{ial} x_{ijkl} \right),$$

$$\beta_i \sim \mathcal{N}(\Theta z_i, \Sigma),$$

Priors:

$$\Theta_{k\ell} \sim \mathcal{N}(0, 5),$$

$$\Sigma \sim \text{diag}(\boldsymbol{\tau}) * I * \text{diag}(\boldsymbol{\tau}),$$

$$\tau_k \sim \text{Cauchy}^+(0, 2), \quad \text{with } 0 < \tau_k < 10.$$

We use $\Theta_{k\ell} \sim \mathcal{N}(0, 5)$ to place a weakly informative prior on the population-level effects of covariates. The scale of 10 assumes that a wide range of effects are possible, allowing the data to dominate the posterior when sufficient information exists. For the respondent-level standard deviations, we specify a half-Cauchy prior with an upper bound of 5 to regularize variation across individuals while still allowing for occasional large deviations. This combination of priors has been

recommended in the hierarchical modeling literature for balancing flexibility and stability during sampling (Gelman et al. 2008, Polson and Scott 2012).

EC.1.6. Estimation Accuracy

We assessed model performance using 5-fold cross-validation and report the hit rate, which is the proportion of correctly predicted choices, for both training and test sets. As shown in Table EC.3, the training hit rate remained stable across folds, while the test hit rate varied between 0.419 and 0.492, indicating modest predictive performance on unseen data.

Table EC.3 Hit Rate from Cross-Validation and Full Sample

Fold	Train Hit Rate	Test Hit Rate
1	0.822	0.419
2	0.821	0.450
3	0.811	0.476
4	0.806	0.492
5	0.807	0.444

We also checked several standard indicators to make sure that our model ran properly and produced reliable results. First, there were no divergent transitions, which means the algorithm did not struggle to explore certain areas of the model. The tree depth was within the expected range, showing that the algorithm was able to take long enough steps to explore the model effectively. In addition, the effective sample sizes were large enough for all parameters, which means the estimates were stable and based on enough independent information. Lastly, all $\hat{R}$ values were close to 1, indicating that the different simulation chains agreed with each other and the model had fully converged. These checks together show that the model fit was successful and the results can be trusted.

EC.1.7. Robustness Test

Table EC.4 shows the estimation results of our analyses excluding participants who answered at least one of the attention check questions wrong and who had never used paid street parking. The results show that our findings are robust to the sample we used in the analysis.

EC.1.8. Further Analysis of Survey Results

To examine how well participants' identify the attribute importance aligned with their actual decision-making behavior, we compared stated rankings with revealed rankings. Stated importance was based on self-reported rankings of the three parking attributes asked at the end of the choice tasks, whereas revealed importance was derived from the range of estimated part-worth utilities per

Table EC.4 Posterior summaries of part-worth estimates under different exclusion criteria.

Attribute Label	Correct Attention Check (n=331)				Removed Recency (n=314)			
	Mean	Std. Dev.	L. 95% CI	U. 95% CI	Mean	Std. Dev.	L. 95% CI	U.95% CI
$\beta_{NoChoice}$	-4.056	0.320	-4.738	-3.477	-3.923	0.316	-4.593	-3.362
$\beta_{Price:3\$/hour}$	1.529	0.061	1.412	1.650	1.539	0.062	1.419	1.661
$\beta_{Price:4\$/hour}$	0.700	0.048	0.606	0.793	0.698	0.049	0.603	0.794
$\beta_{Price:5\$/hour}$	0.158	0.045	0.070	0.246	0.149	0.045	0.062	0.239
$\beta_{Price:6\$/hour}$	-0.500	0.049	-0.596	-0.405	-0.504	0.050	-0.601	-0.405
$\beta_{Distance:50m}$	1.586	0.060	1.471	1.705	1.503	0.060	1.388	1.621
$\beta_{Distance:100m}$	1.025	0.052	0.924	1.127	0.993	0.053	0.891	1.098
$\beta_{Distance:200m}$	0.417	0.048	0.323	0.510	0.389	0.049	0.292	0.484
$\beta_{Distance:400m}$	-0.635	0.051	-0.737	-0.536	-0.614	0.052	-0.716	-0.514
$\beta_{Extension:<20\%}$	1.404	0.059	1.288	1.523	1.352	0.060	1.239	1.470
$\beta_{Extension:20\%-40\%}$	0.958	0.051	0.859	1.059	0.919	0.051	0.822	1.021
$\beta_{Extension:41\%-60\%}$	-0.019	0.045	-0.108	0.068	-0.032	0.046	-0.123	0.057
$\beta_{Extension:61\%-80\%}$	-0.760	0.053	-0.867	-0.657	-0.711	0.054	-0.818	-0.606

attribute. On average, participants tended to overstate the importance of *Parking Price* (mean stated – revealed rank difference: 0.43), while slightly understating the roles of *Distance to Destination* (–0.30) and *Likelihood of Extension* (–0.13). When examining the number of attributes each participant ranked correctly, only 28% matched all three ranks, 45% matched one, and 27% did not match any. In addition, 46 participants (approximately 10% of the sample) reported attribute rankings that were in the exact reverse order of their revealed importance. These discrepancies highlight the value of methods in uncovering latent decision strategies that may not be accessible through self-report alone.

While these results highlight systematic discrepancies between perceived and actual decision weights, they are based on relative rankings alone. To gain a more nuanced understanding of how participants weighted each attribute, we next computed attribute importance scores by calculating the range of part-worth utilities for each attribute and normalizing them across all three attributes for each participant. The resulting values represent the proportion of total utility variation attributable to each attribute. To explore whether the payment method influenced decision strategies, we computed relative attribute importance scores across participants assigned to different payment methods. As shown in Table EC.5, *Parking Price* was the most influential attribute across all groups, which accounts for 46.1% of decision weight for cash users, 45.2% for credit card users, and 49.4% for mobile app users. In contrast, the *Likelihood of Extension* had the lowest relative importance in all groups, contributing only 24.9% for cash users, 25.7% for credit card users, and just 24.0% for mobile app users. This result is particularly noteworthy for participants assigned to mobile application, who placed the least weight on *Likelihood of Extension* and *Distance to Destination* despite placing the most weight on *Parking Price*. This result also coincide with our previous

observations on the effect of convenience of mobile application usage. As mobile applications allow drivers to extend their parking duration remotely, participants assigned to mobile application may perceive uncertain parking durations as less costly or inconvenient than those relying on physical payments.

Table EC.5 Average Relative Attribute Importance by Payment Method

Payment Method	Price	Distance to Destination	Likelihood of Extension
Cash	0.461	0.290	0.249
Credit Card	0.452	0.291	0.257
Mobile Application	0.494	0.265	0.240

EC.2. Modeling Part

In this part, we will provide the necessary further details in the explanation of our mathematical modeling approach. Table EC.6 shows the notation used in our mathematical formulation throughout this study.

Table EC.6 Notation used in Pricing Model

Sets	
N	streets
T	time intervals
M	payment method
Parameters	
N_{it}	estimated number of i -type cars at time t
Cap_j	capacity of street j
d_{ij}	distance between street i and street j
h_m	hassle cost of needing to increase parking payment using payment method m
κ_i	probability of being S -type of driver on street i
$\mu_{i\phi}$	mean payment amount for $i\phi$ -type drivers
β_l	coefficient of parameter l on utility
δ_{mi}	probability of paying using payment method m on street i
lb, ub	lower and upper bound of price
Decision Variables	
$\mathbb{L}_{jt}$	number of spots available on street j at time t
s_{jt}	binary variable 1 if there is availability on street j at time t
$y_{ijtm}^{v\phi}$	the number of ivm -type travellers who attempt to park their cars at node j in period t for ϕ duration
$a_{ijtm}^{v\phi}$	the number of ivm -type travellers who park their cars at node j in period t for ϕ duration
p_j^τ	price of parking on street j for duration τ
le_{ij}	extension probability of i -type drivers on street j
η_{ij}	utility of parking on street i instead of street j at time t
ω_{ij}	probability of parking on street i instead of street j
$\rho_{v\phi}^{ijm}$	the probability of paying ϕ period on street j even if the driver was ivm -type driver

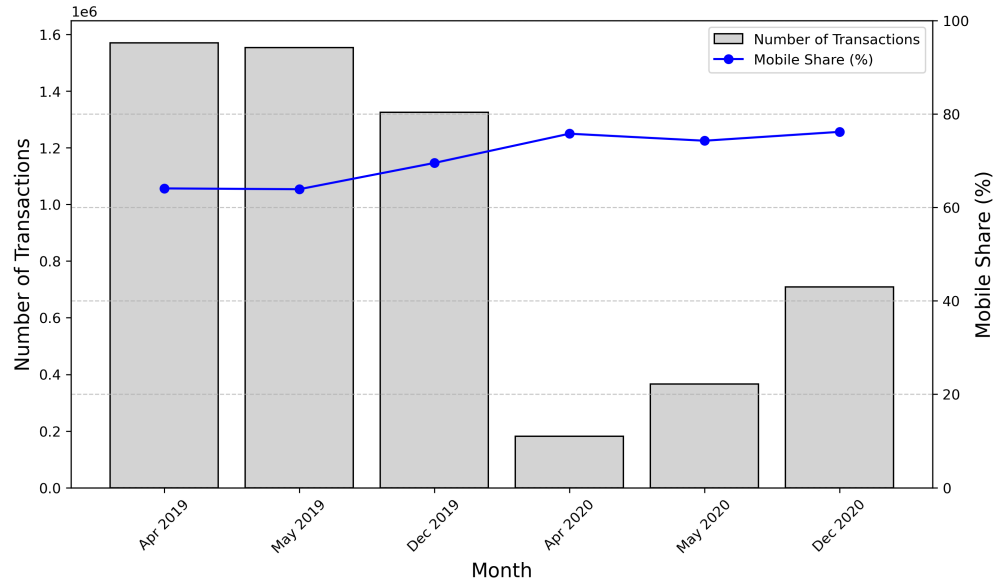
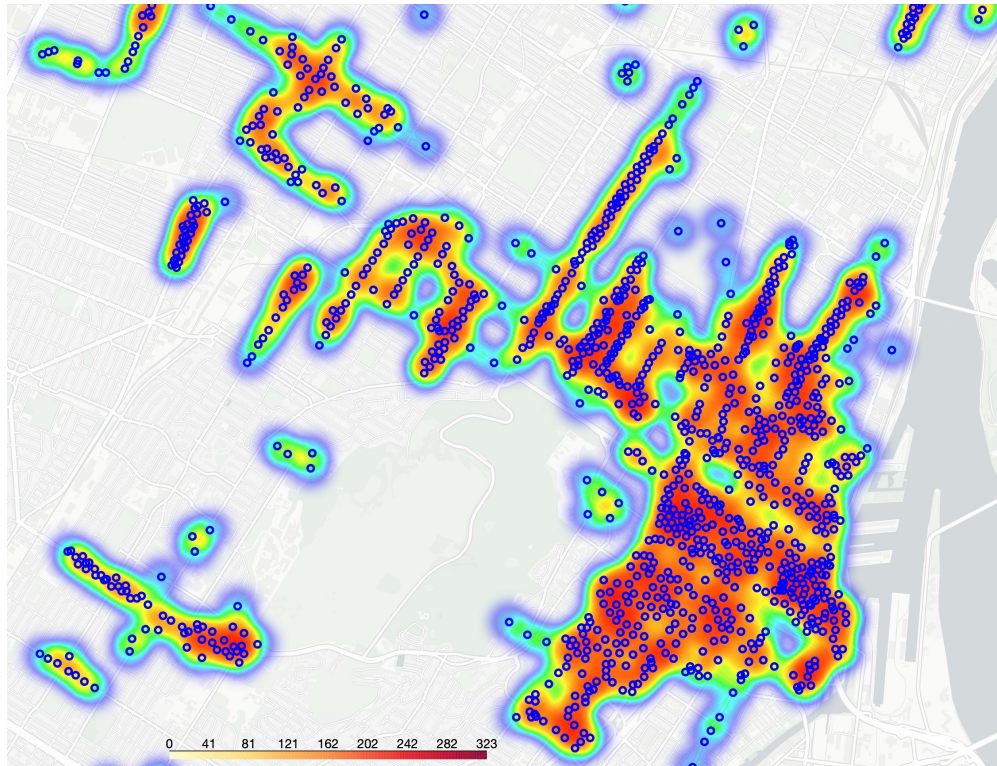
EC.2.1. Discrete Pricing Model

Assuming that prices are selected from a finite set of discrete options, we have the following mixed-integer linear programming model.

$$\begin{aligned} \max \quad & \sum_{t=1}^T \sum_{i=1}^N \sum_{j=1}^N \sum_{b=1}^{PL} \sum_{m \in M} \frac{1}{4} \left[\alpha_b^{S\theta} \theta_{ijtbm}^{LS} \mu_{i,L} + \sum_{\varrho \in \{S,L\}} \alpha_b^{\varrho} (\theta_{ijtbm}^{SS} D_{i,S,\varrho} + \sum_{v \in \{S,L\}} \theta_{ijtbm}^{vL} D_{i,L,\varrho}) \right] \\ \text{s.t.} \quad & \mathbb{L}_{jt} = \mathbb{L}_{jt-1} + \sum_{i=1}^N \sum_{m \in M} \sum_{v, \phi \in \{S,L\}} a_{ijtm}^{v\phi} - \sum_{i=1}^N \sum_{m \in M} \sum_{\phi \in \{S,L\}} a_{ij(t-\mu_{iL})m}^{L\phi} + a_{ij(t-\mu_{i\phi})m}^{S\phi}, \quad \forall j \in N, \forall t \in T \quad (17) \\ & \mathbb{L}_{jt} \leq Cap_j, \quad \forall j \in N, \forall t \in T \quad (18) \\ & s_{jt} \leq Cap_j - \mathbb{L}_{jt}, \quad \forall j \in N, \forall t \in T \quad (19) \\ & Cap_j - \mathbb{L}_{jt} \leq s_{jt} Cap_j, \quad \forall j \in N, \forall t \in T \quad (20) \\ & p_j^\tau = \sum_{b \in PL} \alpha_b^\tau \gamma_{bj}, \quad \forall j \in N, \forall \tau \in \{S, L\} \quad (21) \\ & \sum_{b \in PL} \gamma_{bj} = 1, \quad \forall j \in N \quad (22) \\ & a_{ijtm}^{v\phi} \leq \sum_{b \in PL} \bar{y}_{ijtbm}^{v\phi} \gamma_{bj}, \quad \forall i, j \in N, \forall t \in T, \forall v, \phi \in \{S, L\}, \forall m \in M \quad (23) \\ & a_{ijtm}^{v\phi} - \theta_{ijtbm}^{v\phi} \leq Cap_j (1 - \gamma_{bj}), \quad \forall i, j \in N, \forall t \in T, \forall v, \phi \in \{S, L\}, \forall b \in PL, \forall m \in M \quad (24) \\ & 0 \leq \theta_{ijtbm}^{v\phi} \leq a_{ijtm}^{v\phi}, \quad \forall i, j \in N, \forall t \in T, \forall v, \phi \in \{S, L\}, \forall b \in PL, \forall m \in M \quad (25) \\ & \theta_{ijtbm}^{v\phi} \leq Cap_j \gamma_{bj}, \quad \forall i, j \in N, \forall t \in T, \forall v, \phi \in \{S, L\}, \forall b \in PL, \forall m \in M \quad (26) \\ & \gamma_{bj} \in \{0, 1\}, \quad \forall j \in N, \forall b \in PL \quad (27) \\ & s_{jt} \in \{0, 1\}, \quad \forall j \in N, \forall t \in T \quad (28) \\ & + \text{non-negativity constraints} \end{aligned}$$

EC.2.2. Transaction Data and Descriptive Statistics of Parameters

Figure EC.6 illustrates monthly parking transaction volumes alongside the share of mobile payments, which highlights temporal variation in demand and the steady increase in mobile adoption over time. Furthermore, Figure EC.7 visualizes adjusted parking transaction counts across street intersections to capture spatial variation in curbside activity. Due to visual density and spatial dispersion, it was not possible to display all intersections simultaneously; therefore, we focus only on the regions with high parking activity. Each circle represents a street intersection, with number of transactions normalized by local parking capacity to reflect relative demand intensity. Warmer colors in the underlying heatmap indicate higher adjusted usage intensity, with red zones reflecting intersections that experience disproportionately high parking demand relative to capacity.

Figure EC.6 Monthly Parking Transactions and Mobile Payment Share**Figure EC.7 Transaction Data Heatmap**

In our study, we operationalize the duration structure using the two representative duration values, μ_{iS} and μ_{iL} , and the proportion of short-duration drivers for each street, κ_i . To estimate

these quantities, we use a Gaussian Mixture Model (GMM) approach applied to the distribution of discretized payment durations. We do not estimate κ_i directly from transaction-level payment choices, as these reflect observed behaviors that may be influenced by contextual factors such as pricing or hassle of extension. In particular, a short-type driver may still choose to pay for a long duration due to the inconvenience of returning to the vehicle when using non-mobile payment methods. As a result, using the frequency of short payments as a proxy would shadow true duration preferences with strategic behaviors under uncertainty. Instead, we apply a GMM-based latent variable approach that infers κ_i from the shape of the duration distribution. As a result, for each street i , we fit a GMM, assuming that each transaction y_n is drawn from the distribution:

$$f(y_n) = \kappa_i \mathcal{N}(y_n | \mu_{iS}, \sigma_S^2) + (1 - \kappa_i) \mathcal{N}(y_n | \mu_{iL}, \sigma_L^2). \quad (29)$$

The parameters $\mu_{iS}, \mu_{iL}, \sigma_S, \sigma_L, \kappa_i$ are estimated using the Expectation-Maximization (EM) algorithm implemented in the standard GMM fitting procedure. We interpret the component with the smaller mean as representing short-duration drivers and use its corresponding mixture weight as the estimate for κ_i , while the component means are taken as estimates of μ_{iS} and μ_{iL} . To reduce the risk of convergence failure and having unreliable outputs, we restrict the analysis to streets with at least ten observed transactions. Table EC.7 presents descriptive statistics of duration parameters and mobile payment usage.

Table EC.7 Descriptive Statistics of Duration and Behavioral Parameters

Variable	Mean	Std. Dev.	Min	10%	Median	90%	Max
μ_{iS} (min)	39.78	11.93	15.00	15.00	45.00	45.00	90.00
μ_{iL} (min)	115.33	23.18	60.00	75.00	120.00	135.00	180.00
κ_i	0.62	0.13	0.21	0.46	0.61	0.81	0.98
Mobile Payment Ratio	0.63	0.14	0.21	0.42	0.65	0.78	0.99

EC.2.3. Neighborhood Expansion Algorithm

We group streets based on their physical proximity using the following procedure. The purpose of this grouping is twofold: (1) to reflect substitution effects between nearby streets, and (2) to reduce the computational burden by solving the pricing model over localized subsets of streets rather than the full network. This procedure ensures that each resulting group constitutes a localized area where pricing decisions can be optimized jointly. The detailed procedure is as follows:

Algorithm 1 Neighborhood Expansion with Iteration and Size Constraints

```

1: Initialize an empty set processed_streets
2: Input parameters:
3:   MAX_ITER: maximum number of expansion iterations
4:   DIST_THRESHOLD: distance threshold for neighbor inclusion
5:   MIN_GROUP_SIZE, MAX_GROUP_SIZE: allowable group size bounds
6: for each street s in the list of streets do
7:   if  $s \in \textit{processed\_streets}$  then
8:     continue to the next street
9:   end if
10:   $\textit{to\_explore} \leftarrow \{s\}$ ,  $\textit{all\_neighbors} \leftarrow \{s\}$ ,  $\textit{iter} \leftarrow 0$ 
11:  while  $\textit{to\_explore} \neq \emptyset$  and  $\textit{iter} < \textit{MAX\_ITER}$  do
12:    Initialize  $\textit{new\_neighbors} \leftarrow \emptyset$ 
13:    for each  $t \in \textit{to\_explore}$  do
14:      Identify neighbors of  $t$  within DIST_THRESHOLD not in  $\textit{processed\_streets} \cup$ 
         $\textit{all\_neighbors}$ 
15:      Add these to  $\textit{new\_neighbors}$ 
16:    end for
17:     $\textit{all\_neighbors} \leftarrow \textit{all\_neighbors} \cup \textit{new\_neighbors}$ 
18:     $\textit{to\_explore} \leftarrow \textit{new\_neighbors}$ 
19:     $\textit{iter} \leftarrow \textit{iter} + 1$ 
20:  end while
21:   $\textit{considered\_streets} \leftarrow \textit{sorted}(\textit{all\_neighbors})$ 
22:  if  $|\textit{considered\_streets}| \leq \textit{MIN\_GROUP\_SIZE}$  or  $|\textit{considered\_streets}| \geq \textit{MAX\_GROUP\_SIZE}$ 
    then
23:    continue to the next street
24:  end if
25:   $\textit{processed\_streets} \leftarrow \textit{processed\_streets} \cup \textit{considered\_streets}$ 
26: end for

```

EC.2.4. Computational Study

In this subsection, we provide the complementary information for the observational group-level analysis. Table EC.8 presents the number of neighbors considered in each group, along with the maximum, minimum, and mean values for capacity, short-term parker proportion, and mobile payment proportion.

Table EC.8 Group-level Descriptive Statistics

Group Names	# of Neighbors	Capacity			Proportion of Short-term Parkers			Proportion of Mobile Users		
		Min	Max	Mean	Min	Max	Mean	Min	Max	Mean
Group 1	6	6	15	9.667	0.646	0.851	0.760	0.506	0.718	0.614
Group 2	6	6	26	14.000	0.350	0.854	0.721	0.349	0.443	0.401
Group 3	7	5	21	11.571	0.458	0.832	0.683	0.305	0.423	0.379
Group 4	6	6	14	9.333	0.646	0.855	0.721	0.514	0.712	0.621
Group 5	8	5	21	10.625	0.545	0.658	0.611	0.221	0.339	0.281
Group 6	6	6	49	18.333	0.404	0.497	0.445	0.192	0.251	0.223
Group 7	7	5	10	7.857	0.636	0.854	0.801	0.259	0.362	0.306
Group 8	6	6	20	12.000	0.445	0.524	0.483	0.262	0.371	0.310
Group 9	8	7	30	16.625	0.535	0.801	0.676	0.393	0.485	0.422
Group 10	6	12	36	21.833	0.566	0.632	0.604	0.237	0.270	0.251
Group 11	7	5	39	19.857	0.442	0.584	0.528	0.169	0.282	0.224

Table EC.9 summarizes how streets were grouped into categorical bins for analysis based on three key variables. Each variable is divided into three quantile-based categories. *Bin Edges* column shows the numerical thresholds defining the range of each category. Finally, *Counts* column indicates the number of streets falling into each bin, with roughly equal distribution across groups due to quantile classification.

Table EC.9 Number of Streets in Each Variable Group

Variable	Bin Edges	Counts
Capacity	[5.0, 9.0, 15.0, 49.0]	Low: 31, Medium: 21, High: 21
δ_{mi}	[0.17, 0.28, 0.42, 0.72]	Low: 25, Medium: 24, High: 24
$\mu_i^{combined}$	[2.46, 4.13, 5.23, 5.98]	Short: 25, Medium: 24, Long: 24